Theory of Algebraic Topology 15

July 29, 2026

Lecture notes with worked examples in algebraic topology

Contents

1 Problem 1

Exact statement

(a) Show that there is no orientation-reversing homeomorphism

$$f : \mathbb{CP}^2 \longrightarrow \mathbb{CP}^2.$$

(b) If

$$f : S^2 \times S^2 \longrightarrow \mathbb{CP}^2,$$

show that

$$f_* : H_4(S^2 \times S^2) \longrightarrow H_4(\mathbb{CP}^2)$$

has even degree.

Solution

Part (a): No orientation-reversing homeomorphism of $\mathbb{CP}^2$

We use the cohomology ring of complex projective space:

$$H^*(\mathbb{CP}^2; \mathbb{Z}) \cong \mathbb{Z}[\alpha]/(\alpha^3),$$

where

$$\alpha \in H^2(\mathbb{CP}^2; \mathbb{Z})$$

is a generator, and

$$\alpha^2 \in H^4(\mathbb{CP}^2; \mathbb{Z})$$

is a generator of the top cohomology group.
Let

$$f : \mathbb{CP}^2 \longrightarrow \mathbb{CP}^2$$

be a homeomorphism. Since

$$H^2(\mathbb{CP}^2; \mathbb{Z}) \cong \mathbb{Z},$$

the induced map

$$f^* : H^2(\mathbb{CP}^2; \mathbb{Z}) \longrightarrow H^2(\mathbb{CP}^2; \mathbb{Z})$$

must be multiplication by either 1 or -1. Hence

$$f^*(\alpha) = \pm\alpha.$$

Now consider the induced map on the top-dimensional cohomology class

$$\alpha^2 \in H^4(\mathbb{CP}^2; \mathbb{Z}).$$

Because f^* preserves cup products, we have

$$f^*(\alpha^2) = f^*(\alpha \smile \alpha) = f^*(\alpha) \smile f^*(\alpha).$$

Therefore
$$f^*(\alpha^2) = (\pm\alpha) \smile (\pm\alpha) = \alpha^2.$$

So every homeomorphism of $\mathbb{CP}^2$ acts trivially on
$$H^4(\mathbb{CP}^2; \mathbb{Z}).$$

However, if f were orientation-reversing, then f^* would act by multiplication by -1 on the top cohomology group. Thus we would have
$$f^*(\alpha^2) = -\alpha^2.$$

This contradicts the previous computation
$$f^*(\alpha^2) = \alpha^2.$$

Hence no orientation-reversing homeomorphism
$$f : \mathbb{CP}^2 \longrightarrow \mathbb{CP}^2$$
exists.

Therefore,
$$\boxed{\mathbb{CP}^2 \text{ admits no orientation-reversing homeomorphism.}}$$

Part (b): A map $S^2 \times S^2 \to \mathbb{CP}^2$ has even degree

Let
$$u \in H^2(\mathbb{CP}^2; \mathbb{Z})$$
be a generator. Then
$$u^2 \in H^4(\mathbb{CP}^2; \mathbb{Z})$$
is a generator of the top cohomology group.

Now consider the cohomology ring of $S^2 \times S^2$. Let
$$a, b \in H^2(S^2 \times S^2; \mathbb{Z})$$
be the pullbacks of the fundamental cohomology classes from the two sphere factors. Then
$$H^*(S^2 \times S^2; \mathbb{Z}) \cong \mathbb{Z}[a, b]/(a^2, b^2),$$
where
$$ab \in H^4(S^2 \times S^2; \mathbb{Z})$$
is a generator.

Since
$$H^2(S^2 \times S^2; \mathbb{Z}) \cong \mathbb{Z} \oplus \mathbb{Z},$$
we may write
$$f^*(u) = ma + nb$$

for some integers $m, n \in \mathbb{Z}$.

Now compute the pullback of the top-dimensional generator u^2. Since f^* preserves cup products,

$$f^*(u^2) = f^*(u)^2.$$

Thus

$$f^*(u^2) = (ma + nb)^2.$$

Expanding, we obtain

$$(ma + nb)^2 = m^2a^2 + 2mnab + n^2b^2.$$

But in the cohomology ring of $S^2 \times S^2$, we have

$$a^2 = 0, \qquad b^2 = 0.$$

Therefore

$$f^*(u^2) = 2mnab.$$

The degree of f is the integer d such that

$$f^*(u^2) = d\,ab.$$

Comparing this with

$$f^*(u^2) = 2mnab,$$

we get

$$d = 2mn.$$

Hence d is even.

Therefore,

$$\boxed{\deg(f) \text{ is even.}}$$

Equivalently,

$$f_*[S^2 \times S^2] = 2mn[\mathbb{CP}^2],$$

so the induced homomorphism

$$f_* : H_4(S^2 \times S^2) \longrightarrow H_4(\mathbb{CP}^2)$$

has even degree.

2 Problem 2

Exact statement

Suppose M_1 and M_2 are closed, connected, oriented n-manifolds and $x_i \in M_i$. Choose charts

$$\phi_i : U_i \longrightarrow \mathbb{R}^n,$$

where

$$x_i \in U_i \subset M_i, \qquad \phi_i(x_i) = 0,$$

with the property that

$$\phi_{1*}([M_1]|_{x_1}) = -\phi_{2*}([M_2]|_{x_2}).$$

Let

$$M_i' = M_i - \phi_i^{-1}(B_1(0)),$$

where $B_1(0)$ is the open ball of radius 1. Define the connected sum

$$M_1 \# M_2 := M_1' \cup M_2' / \sim,$$

where

$$\phi_1^{-1}(x) \sim \phi_2^{-1}(x) \qquad \text{for } x \in S^{n-1}.$$

(a) Explain why $M_1 \# M_2$ is a manifold.

(b) Show that

$$H^i(M_1 \# M_2) \cong H^i(M_1) \oplus H^i(M_2)$$

for $0 < i < n$.

(c) Show that there is an orientation $[M_1 \# M_2]$ on $M_1 \# M_2$ such that

$$[M_1 \# M_2]|_{M_i'} = [M_i]|_{M_i'}$$

for $i = 1, 2$.

(d) Show that the cup product pairing on $H^*(M_1 \# M_2)$ is given by

$$((a_1, a_2), (b_1, b_2))_\# = (a_1, b_1)_1 + (a_2, b_2)_2,$$

for

$$0 < |a_1| = |a_2| < n, \qquad 0 < |b_1| = |b_2| < n,$$

where $(\cdot, \cdot)_i$ is the cup product pairing on M_i.

Solution.

Part (a): Why $M_1 \# M_2$ is a manifold

The connected sum is constructed by deleting an open n-ball from each manifold and then gluing the resulting boundary spheres.

For each $i = 1, 2$, the space

$$M_i' = M_i - \phi_i^{-1}(B_1(0))$$

is an n-manifold with boundary. Its boundary is

$$\partial M_i' = \phi_i^{-1}(S^{n-1}).$$

Away from the boundary, nothing changes. Every point in the interior of M_1' or M_2' already has a neighborhood homeomorphic to an open subset of $\mathbb{R}^n$.

Thus the only point to check is what happens near the glued sphere. Near a boundary point, each M_i' looks locally like a half-space:

$$\mathbb{R}^{n-1} \times [0, \varepsilon).$$

When two such half-spaces are glued along their common boundary

$$\mathbb{R}^{n-1} \times \{0\},$$

the result is locally

$$\mathbb{R}^{n-1} \times (-\varepsilon, \varepsilon) \cong \mathbb{R}^n.$$

Therefore every point of the glued space has a neighborhood homeomorphic to $\mathbb{R}^n$. Hence

$$\boxed{M_1 \# M_2 \text{ is an } n\text{-manifold.}}$$

Since M_1 and M_2 are closed and connected, the connected sum is also closed and connected.

Part (b): Cohomology of the connected sum

Let

$$X = M_1 \# M_2.$$

We cover X by two open sets A and B, where A is a small open thickening of M_1', and B is a small open thickening of M_2'. Then

$$A \simeq M_1', \qquad B \simeq M_2',$$

and

$$A \cap B \simeq S^{n-1}.$$

Now use the Mayer–Vietoris sequence:

$$\cdots \longrightarrow H^{i-1}(A \cap B) \longrightarrow H^i(X) \longrightarrow H^i(A) \oplus H^i(B) \longrightarrow H^i(A \cap B) \longrightarrow \cdots.$$

Since

$$A \cap B \simeq S^{n-1},$$

we have

$$H^i(A \cap B) = 0$$

for

$$0 < i < n - 1.$$

Therefore, for $0 < i < n - 1$, the Mayer–Vietoris sequence gives

$$H^i(X) \cong H^i(A) \oplus H^i(B).$$

Because

$$A \simeq M_1', \qquad B \simeq M_2',$$

we obtain

$$H^i(X) \cong H^i(M_1') \oplus H^i(M_2').$$

Removing an open n-ball from a connected closed n-manifold does not change cohomology in degrees $0 < i < n$. Hence

$$H^i(M_j') \cong H^i(M_j)$$

for $j = 1, 2$. Thus

$$H^i(M_1 \# M_2) \cong H^i(M_1) \oplus H^i(M_2)$$

for $0 < i < n - 1$.

It remains to discuss the degree $i = n - 1$. The relevant part of the Mayer–Vietoris sequence is

$$0 \longrightarrow H^{n-1}(X) \longrightarrow H^{n-1}(M_1') \oplus H^{n-1}(M_2') \longrightarrow H^{n-1}(S^{n-1}) \longrightarrow H^n(X) \longrightarrow 0.$$

Since X is a closed connected oriented n-manifold,

$$H^n(X) \cong \mathbb{Z}.$$

The connecting map

$$H^{n-1}(S^{n-1}) \longrightarrow H^n(X)$$

identifies the generator of the gluing sphere with the top orientation class of X. Hence this map is an isomorphism. Therefore the preceding map

$$H^{n-1}(M_1') \oplus H^{n-1}(M_2') \longrightarrow H^{n-1}(S^{n-1})$$

has zero image, and we get

$$H^{n-1}(X) \cong H^{n-1}(M_1') \oplus H^{n-1}(M_2').$$

Again,

$$H^{n-1}(M_j') \cong H^{n-1}(M_j),$$

so

$$H^{n-1}(M_1 \# M_2) \cong H^{n-1}(M_1) \oplus H^{n-1}(M_2).$$

Therefore, for every $0 < i < n$,

$$\boxed{H^i(M_1 \# M_2) \cong H^i(M_1) \oplus H^i(M_2).}$$

Part (c): Orientation of the connected sum

Each M_i' inherits an orientation from M_i. The only issue is whether the two orientations agree along the glued boundary sphere.

The assumption

$$\phi_{1*}([M_1]|_{x_1}) = -\phi_{2*}([M_2]|_{x_2})$$

means that the two coordinate balls are identified with opposite orientations.

This is exactly the condition needed when forming the connected sum of oriented manifolds. Indeed, when an open ball is removed from an oriented manifold, the induced boundary orientation on the resulting boundary sphere is determined by the outward normal convention. The boundary orientations on

$$\partial M_1' \qquad \text{and} \qquad \partial M_2'$$

are opposite under the gluing map. Hence the two orientations fit together after gluing.

Therefore there exists an orientation class

$$[M_1 \# M_2] \in H_n(M_1 \# M_2)$$

such that

$$\boxed{[M_1 \# M_2]|_{M_i'} = [M_i]|_{M_i'}}$$

for $i = 1, 2$.

Thus the connected sum orientation restricts to the original orientations on both punctured pieces.

Part (d): Cup product pairing on the connected sum

By part (b), for $0 < i < n$, we have

$$H^i(M_1 \# M_2) \cong H^i(M_1) \oplus H^i(M_2).$$

Thus a cohomology class on $M_1 \# M_2$ may be written as a pair

$$(a_1, a_2),$$

where

$$a_i \in H^*(M_i).$$

Similarly, write another class as

$$(b_1, b_2).$$

The cup product pairing on $M_1 \# M_2$ is defined by

$$((a_1, a_2), (b_1, b_2))_{\#} = \langle (a_1, a_2) \smile (b_1, b_2), [M_1 \# M_2] \rangle .$$

Under the splitting of cohomology from part (b), the cup product is computed componentwise:

$$(a_1, a_2) \smile (b_1, b_2) = (a_1 \smile b_1,\ a_2 \smile b_2).$$

There are no cross terms such as $a_1 \smile b_2$ or $a_2 \smile b_1$, because these classes come from different sides of the connected sum neck.

Using the orientation from part (c), we have

$$[M_1 \# M_2]|_{M_i'} = [M_i]|_{M_i'}.$$

Therefore evaluation on the fundamental class splits as the sum of the evaluations on the two summands:

$$((a_1, a_2), (b_1, b_2))_\# = \langle a_1 \smile b_1, [M_1] \rangle + \langle a_2 \smile b_2, [M_2] \rangle .$$

By definition,

$$(a_i, b_i)_i = \langle a_i \smile b_i, [M_i] \rangle .$$

Hence

$$\boxed{((a_1, a_2), (b_1, b_2))_\# = (a_1, b_1)_1 + (a_2, b_2)_2.}$$

Thus the cup product pairing on the connected sum is the orthogonal direct sum of the cup product pairings on M_1 and M_2.

3 Theory background for Problem 2: connected sums, orientations, and cup products

3.1 Connected sum of two closed manifolds

Let M_1 and M_2 be closed connected n-manifolds. The connected sum

$$M_1 \# M_2$$

is obtained by removing an open n-ball from each manifold and then gluing the two resulting boundary spheres.

More precisely, choose embedded open balls

$$B_i \subset M_i$$

around points $x_i \in M_i$. Then define

$$M_i' = M_i - B_i.$$

Each M_i' is an n-manifold with boundary

$$\partial M_i' \cong S^{n-1}.$$

The connected sum is then

$$M_1 \# M_2 = M_1' \cup_\partial M_2',$$

where the two boundary spheres are identified by a homeomorphism

$$\partial M_1' \longrightarrow \partial M_2'.$$

Intuitively, one cuts a small disk out of each manifold and joins the two remaining boundary spheres by a neck.

$$\boxed{\text{connected sum} = \text{remove two balls and glue along the resulting spheres}}$$

3.2 Local model near the gluing sphere

The main reason that $M_1 \# M_2$ is again a manifold is that the local gluing model is very simple.

A point in the interior of M_i' already has a neighborhood homeomorphic to $\mathbb{R}^n$. The only new points are points on the glued boundary sphere.

Near a boundary point, an n-manifold with boundary looks like a half-space:

$$\mathbb{R}^{n-1} \times [0, \varepsilon).$$

When two such half-spaces are glued along the boundary

$$\mathbb{R}^{n-1} \times \{0\},$$

we obtain

$$\mathbb{R}^{n-1} \times (-\varepsilon, \varepsilon) \cong \mathbb{R}^n.$$

Thus the gluing removes the boundary and produces ordinary manifold neighborhoods.

Therefore,

$$M_1 \# M_2$$

is an n-manifold without boundary.

3.3 Orientation and why opposite orientations are needed

Suppose now that M_1 and M_2 are oriented.

If M is an oriented n-manifold with boundary, then its boundary ∂M receives the induced boundary orientation. The convention is:

$$(v_1, \dots, v_{n-1}) \text{ is positively oriented in } \partial M$$

if and only if

$$(\nu, v_1, \dots, v_{n-1})$$

is positively oriented in M, where ν is the outward-pointing normal vector.

Equivalently,

$$\text{outward normal first}$$

determines the boundary orientation.

When forming the connected sum of oriented manifolds, the two boundary spheres must be glued with opposite boundary orientations. Otherwise the orientations would not fit together across the neck.

In the problem, the condition

$$\phi_{1*}([M_1]|_{x_1}) = -\phi_{2*}([M_2]|_{x_2})$$

means exactly that the two coordinate balls are identified with opposite orientations.

Therefore the orientations on M_1' and M_2' glue together to give an orientation on

$$M_1 \# M_2.$$

The resulting orientation class satisfies

$$[M_1 \# M_2]|_{M_i'} = [M_i]|_{M_i'}$$

for $i = 1, 2$.

$\boxed{\text{The connected sum of oriented manifolds uses an orientation-reversing boundary identification.}}$

3.4 Cohomology of a punctured manifold

Let M be a closed connected oriented n-manifold, and let

$$M' = M - \operatorname{int}(D^n)$$

be the manifold obtained by deleting an open n-disk.

Then M' is an n-manifold with boundary

$$\partial M' \cong S^{n-1}.$$

Removing an open disk does not change the cohomology in the degrees

$$0 < i < n.$$

Thus

$$H^i(M') \cong H^i(M)$$

for

$$0 < i < n.$$

The only cohomology group that changes is the top cohomology group:

$$H^n(M) \cong \mathbb{Z}, \qquad H^n(M') = 0.$$

This happens because M is closed, but M' has boundary.

So the useful rule is:

$$\boxed{H^i(M - \operatorname{int}(D^n)) \cong H^i(M) \quad \text{for } 0 < i < n.}$$

3.5 Mayer–Vietoris sequence for the connected sum

Let
$$X = M_1 \# M_2.$$
We cover X by two open sets A and B, where A is a small thickening of M_1', and B is a small thickening of M_2'. Then
$$A \simeq M_1', \qquad B \simeq M_2',$$
and
$$A \cap B \simeq S^{n-1}.$$
The Mayer–Vietoris sequence gives
$$\cdots \longrightarrow H^{i-1}(A \cap B) \longrightarrow H^i(X) \longrightarrow H^i(A) \oplus H^i(B) \longrightarrow H^i(A \cap B) \longrightarrow \cdots.$$
Since
$$A \cap B \simeq S^{n-1},$$
we have
$$H^i(A \cap B) = 0$$
for
$$0 < i < n-1.$$
Hence, for $0 < i < n-1$,
$$H^i(X) \cong H^i(A) \oplus H^i(B).$$
Using
$$A \simeq M_1', \qquad B \simeq M_2',$$
and
$$H^i(M_j') \cong H^i(M_j),$$
we get
$$H^i(M_1 \# M_2) \cong H^i(M_1) \oplus H^i(M_2)$$
for $0 < i < n-1$.

The degree $i = n-1$ is slightly more delicate because
$$H^{n-1}(S^{n-1}) \cong \mathbb{Z}.$$
The relevant part of the Mayer–Vietoris sequence is
$$H^{n-1}(M_1 \# M_2) \longrightarrow H^{n-1}(M_1') \oplus H^{n-1}(M_2') \longrightarrow H^{n-1}(S^{n-1}) \longrightarrow H^n(M_1 \# M_2).$$
Since $M_1 \# M_2$ is a closed connected oriented n-manifold,
$$H^n(M_1 \# M_2) \cong \mathbb{Z}.$$

The connecting map
$$H^{n-1}(S^{n-1}) \longrightarrow H^n(M_1 \# M_2)$$
is an isomorphism. Hence the previous map has zero image, and one obtains
$$H^{n-1}(M_1 \# M_2) \cong H^{n-1}(M_1') \oplus H^{n-1}(M_2').$$

Therefore,
$$H^{n-1}(M_1 \# M_2) \cong H^{n-1}(M_1) \oplus H^{n-1}(M_2).$$

Combining all degrees gives
$$\boxed{H^i(M_1 \# M_2) \cong H^i(M_1) \oplus H^i(M_2) \quad \text{for } 0 < i < n.}$$

3.6 Why the statement excludes degrees 0 and n

The formula
$$H^i(M_1 \# M_2) \cong H^i(M_1) \oplus H^i(M_2)$$
is true only for
$$0 < i < n.$$

It fails in degree 0. Since M_1, M_2, and $M_1 \# M_2$ are connected, we have
$$H^0(M_1 \# M_2) \cong \mathbb{Z},$$
but
$$H^0(M_1) \oplus H^0(M_2) \cong \mathbb{Z} \oplus \mathbb{Z}.$$

It also fails in degree n. Since all three manifolds are closed connected oriented n-manifolds,
$$H^n(M_1 \# M_2) \cong \mathbb{Z},$$
but
$$H^n(M_1) \oplus H^n(M_2) \cong \mathbb{Z} \oplus \mathbb{Z}.$$

Thus the connected sum behaves additively only in the middle cohomology degrees.

$$\boxed{\text{Middle cohomology adds, but degree 0 and degree } n \text{ do not.}}$$

3.7 Cup product pairing

Let M be a closed oriented n-manifold. If
$$a \in H^k(M), \qquad b \in H^{n-k}(M),$$
then the cup product pairing is defined by
$$(a, b)_M = \langle a \smile b, [M] \rangle .$$

Here

$$[M] \in H_n(M)$$

is the fundamental class.

Thus the pairing takes two cohomology classes whose degrees add to n, cups them together, and evaluates the result on the fundamental class.

$$\boxed{(a, b)_M = \langle a \smile b, [M] \rangle}$$

3.8 Cup product pairing on a connected sum

For the connected sum, in degrees $0 < i < n$, we have

$$H^i(M_1 \# M_2) \cong H^i(M_1) \oplus H^i(M_2).$$

So a class in the middle cohomology of $M_1 \# M_2$ can be written as

$$(a_1, a_2),$$

where

$$a_i \in H^*(M_i).$$

Similarly, another class can be written as

$$(b_1, b_2).$$

The cup product behaves componentwise:

$$(a_1, a_2) \smile (b_1, b_2) = (a_1 \smile b_1,\ a_2 \smile b_2).$$

There are no cross terms such as

$$a_1 \smile b_2 \qquad \text{or} \qquad a_2 \smile b_1.$$

Geometrically, this is because classes from M_1 and classes from M_2 live on different sides of the connected sum neck.

Using the orientation compatibility

$$[M_1 \# M_2]|_{M_i'} = [M_i]|_{M_i'},$$

we obtain

$$\begin{aligned} ((a_1, a_2), (b_1, b_2))_\# &= \langle (a_1, a_2) \smile (b_1, b_2), [M_1 \# M_2] \rangle \\ &= \langle a_1 \smile b_1, [M_1] \rangle + \langle a_2 \smile b_2, [M_2] \rangle . \end{aligned}$$

Therefore

$$\boxed{((a_1, a_2), (b_1, b_2))_\# = (a_1, b_1)_1 + (a_2, b_2)_2.}$$

Thus the cup product pairing of a connected sum is the orthogonal direct sum of the cup product pairings of the two summands.

3.9 Summary table

Object	Meaning	Role in connected sum
M_i'	$M_i - \mathrm{int}(D^n)$	punctured manifold
$\partial M_i'$	S^{n-1}	sphere along which we glue
$M_1 \# M_2$	$M_1' \cup_\partial M_2'$	connected sum
$H^i(M_1 \# M_2)$	middle cohomology	$H^i(M_1) \oplus H^i(M_2)$
$[M_1 \# M_2]$	fundamental class	restricts to $[M_i]$ on M_i'
$(\cdot,\cdot)_\#$	cup product pairing	$(\cdot,\cdot)_1 \oplus (\cdot,\cdot)_2$

4 Examples for Problem 2

4.1 Example 1: $S^n \# M \cong M$ (Sn#M = M)

Let M be a closed connected oriented n-manifold. Then

$$S^n \# M \cong M.$$

Indeed, removing an open n-disk from S^n leaves a closed n-disk:

$$S^n - \mathrm{int}(D^n) \cong D^n.$$

Therefore

$$S^n \# M = D^n \cup_{S^{n-1}} \Big(M - \mathrm{int}(D^n)\Big).$$

But gluing D^n back along the boundary sphere exactly fills the hole in M. Hence

$$\boxed{S^n \# M \cong M.}$$

Thus S^n acts as the identity element for connected sum.

4.2 Example 2: $S^n \# S^n \cong S^n$

Take

$$M_1 = S^n, \qquad M_2 = S^n.$$

Since

$$S^n - \mathrm{int}(D^n) \cong D^n,$$

we get

$$S^n \# S^n = D^n \cup_{S^{n-1}} D^n.$$

Two n-disks glued along their common boundary sphere form an n-sphere:

$$D^n \cup_{S^{n-1}} D^n \cong S^n.$$

Therefore

$$\boxed{S^n \# S^n \cong S^n.}$$

This example also explains why the cohomology formula for connected sums does not hold in degree n. We have

$$H^n(S^n \# S^n) \cong H^n(S^n) \cong \mathbb{Z},$$

but

$$H^n(S^n) \oplus H^n(S^n) \cong \mathbb{Z} \oplus \mathbb{Z}.$$

Thus the formula

$$H^i(M_1 \# M_2) \cong H^i(M_1) \oplus H^i(M_2)$$

is only valid for

$$0 < i < n.$$

4.3 Example 3: Connected sum of two tori

Let

$$M_1 = T^2, \qquad M_2 = T^2.$$

Then

$$T^2 \# T^2$$

is a closed orientable surface of genus 2. Geometrically, one removes one open disk from each torus and glues the two resulting boundary circles. The result is a surface with two handles.

Thus

$$\boxed{T^2 \# T^2 \cong \Sigma_2,}$$

where Σ_2 denotes the closed orientable surface of genus 2.

By the connected sum cohomology formula,

$$H^1(T^2 \# T^2) \cong H^1(T^2) \oplus H^1(T^2).$$

Since

$$H^1(T^2) \cong \mathbb{Z}^2,$$

we obtain

$$H^1(T^2 \# T^2) \cong \mathbb{Z}^2 \oplus \mathbb{Z}^2 \cong \mathbb{Z}^4.$$

This agrees with the known formula

$$H^1(\Sigma_g) \cong \mathbb{Z}^{2g}.$$

For $g = 2$, this gives

$$H^1(\Sigma_2) \cong \mathbb{Z}^4.$$

4.4 Example 4: General orientable surfaces

Let Σ_g denote the closed orientable surface of genus g. Then

$$\Sigma_g \# \Sigma_h \cong \Sigma_{g+h}.$$

For example,

$$\Sigma_2 \# \Sigma_3 \cong \Sigma_5.$$

The first cohomology of Σ_g is

$$H^1(\Sigma_g) \cong \mathbb{Z}^{2g}.$$

Therefore

$$\begin{aligned} H^1(\Sigma_g \# \Sigma_h) &\cong H^1(\Sigma_g) \oplus H^1(\Sigma_h) \\ &\cong \mathbb{Z}^{2g} \oplus \mathbb{Z}^{2h} \\ &\cong \mathbb{Z}^{2(g+h)}. \end{aligned}$$

This agrees with the fact that

$$\Sigma_g \# \Sigma_h \cong \Sigma_{g+h}.$$

Thus connected sum of orientable surfaces corresponds geometrically to adding handles.

4.5 Example 5: Cup product on $T^2 \# T^2$

Let

$$X = T^2 \# T^2.$$

Choose generators

$$a_1, b_1 \in H^1(T^2)$$

from the first torus, and generators

$$a_2, b_2 \in H^1(T^2)$$

from the second torus.

Then

$$H^1(X) \cong \langle a_1, b_1 \rangle \oplus \langle a_2, b_2 \rangle.$$

The cup product pairing satisfies

$$(a_1, b_1)_\# = 1, \qquad (a_2, b_2)_\# = 1.$$

The cross terms vanish:

$$(a_1, b_2)_\# = 0, \qquad (a_2, b_1)_\# = 0.$$

Hence the cup product pairing matrix, with respect to the ordered basis

$$a_1, b_1, a_2, b_2,$$

is

$$\begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{pmatrix}.$$

This matrix is block diagonal:

$$Q_{T^2\#T^2} = Q_{T^2} \oplus Q_{T^2}.$$

Thus

$$\boxed{Q_{T^2\#T^2} = Q_{T^2} \oplus Q_{T^2}.}$$

This is an explicit example of the rule that the cup product pairing on a connected sum splits as the direct sum of the pairings on the two summands.

4.6 Example 6: $\mathbb{CP}^2\#\mathbb{CP}^2$

Let

$$M_1 = \mathbb{CP}^2, \qquad M_2 = \mathbb{CP}^2.$$

Then

$$X = \mathbb{CP}^2\#\mathbb{CP}^2$$

is a closed oriented 4-manifold.

We know that

$$H^2(\mathbb{CP}^2) \cong \mathbb{Z},$$

generated by a class u, and

$$\langle u^2, [\mathbb{CP}^2]\rangle = 1.$$

Therefore, by the connected sum cohomology formula,

$$\begin{aligned} H^2(\mathbb{CP}^2\#\mathbb{CP}^2) &\cong H^2(\mathbb{CP}^2) \oplus H^2(\mathbb{CP}^2) \\ &\cong \mathbb{Z} \oplus \mathbb{Z}. \end{aligned}$$

Let

$$u_1 = (u, 0), \qquad u_2 = (0, u).$$

Then the cup product pairing satisfies

$$u_1^2 = 1, \qquad u_2^2 = 1, \qquad u_1u_2 = 0.$$

Thus the intersection form is

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Therefore

$$\boxed{Q_{\mathbb{CP}^2\#\mathbb{CP}^2} = \langle 1\rangle \oplus \langle 1\rangle.}$$

4.7 Example 7: $\mathbb{CP}^2 \# \overline{\mathbb{CP}}^2$

Let

$$\overline{\mathbb{CP}}^2$$

denote $\mathbb{CP}^2$ with the opposite orientation. Consider

$$X = \mathbb{CP}^2 \# \overline{\mathbb{CP}}^2.$$

Again,

$$H^2(X) \cong \mathbb{Z} \oplus \mathbb{Z}.$$

On $\mathbb{CP}^2$, the generator $u \in H^2(\mathbb{CP}^2)$ satisfies

$$\langle u^2, [\mathbb{CP}^2] \rangle = 1.$$

On $\overline{\mathbb{CP}}^2$, the underlying cohomology group is the same, but the fundamental class changes sign. Therefore

$$\langle u^2, [\overline{\mathbb{CP}}^2] \rangle = -1.$$

Hence the cup product pairing on X has matrix

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Thus

$$\boxed{Q_{\mathbb{CP}^2 \# \overline{\mathbb{CP}}^2} = \langle 1 \rangle \oplus \langle -1 \rangle.}$$

This is one of the basic examples in the topology of closed oriented 4-manifolds.

4.8 Example 8: $(S^2 \times S^2) \# \mathbb{CP}^2$

Let

$$M_1 = S^2 \times S^2, \qquad M_2 = \mathbb{CP}^2.$$

Define

$$X = (S^2 \times S^2) \# \mathbb{CP}^2.$$

We have

$$H^2(S^2 \times S^2) \cong \mathbb{Z} \oplus \mathbb{Z}.$$

Let $a, b \in H^2(S^2 \times S^2)$ be the generators coming from the two sphere factors. Then

$$a^2 = 0, \qquad b^2 = 0, \qquad \langle ab, [S^2 \times S^2] \rangle = 1.$$

Also,

$$H^2(\mathbb{CP}^2) \cong \mathbb{Z},$$

with generator u, satisfying
$$\langle u^2, [\mathbb{CP}^2]\rangle = 1.$$

Therefore
$$\begin{aligned} H^2(X) &\cong H^2(S^2 \times S^2) \oplus H^2(\mathbb{CP}^2) \\ &\cong \mathbb{Z}^2 \oplus \mathbb{Z} \\ &\cong \mathbb{Z}^3. \end{aligned}$$

With respect to the basis
$$a, b, u,$$
the cup product pairing matrix is
$$\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Thus
$$\boxed{Q_{(S^2\times S^2)\#\mathbb{CP}^2} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \oplus \langle 1\rangle.}$$

Again, the connected sum gives the orthogonal direct sum of the cup product pairings.

4.9 Summary chart

Connected sum	**Resulting space**	**Middle cohomology / pairing**
$S^n \# M$	M	S^n is the identity for connected sum
$S^n \# S^n$	S^n	degrees 0 and n are not additive
$T^2 \# T^2$	Σ_2	$H^1 \cong \mathbb{Z}^4$
$\Sigma_g \# \Sigma_h$	Σ_{g+h}	$H^1 \cong \mathbb{Z}^{2(g+h)}$
$\mathbb{CP}^2 \# \mathbb{CP}^2$	4-manifold	$\langle 1\rangle \oplus \langle 1\rangle$
$\mathbb{CP}^2 \# \overline{\mathbb{CP}}^2$	4-manifold	$\langle 1\rangle \oplus \langle -1\rangle$
$(S^2 \times S^2) \# \mathbb{CP}^2$	4-manifold	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \oplus \langle 1\rangle$

4.10 Main conclusion

The examples illustrate the main principle:

$$\boxed{\text{Connected sum adds middle cohomology and adds cup product pairings blockwise.}}$$

In degrees 0 and n, the direct sum formula fails because the connected sum is still connected and has only one fundamental class. In the middle degrees, however, the two summands contribute independently.

5 Charts and diagrams for Problem 2

5.1 Diagram 1: Construction of the connected sum

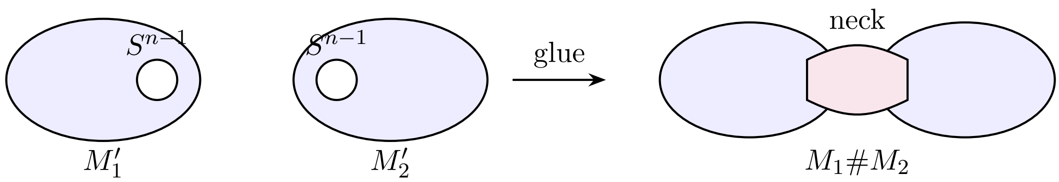


Figure 1: Connected sum: remove one open n-ball from each manifold and glue the two boundary spheres.

5.2 Diagram 2: Local model near the glued sphere

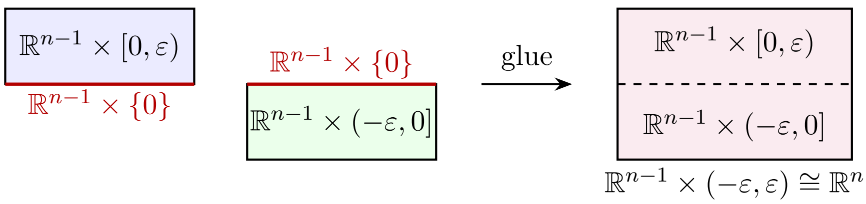


Figure 2: Near the gluing sphere, two boundary half-neighborhoods glue to an ordinary n-dimensional neighborhood.

5.3 Diagram 3: Orientation compatibility

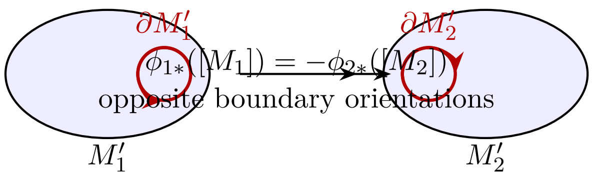


Figure 3: The boundary orientations must be opposite so that the orientations glue correctly across the neck.

5.4 Diagram 4: Mayer–Vietoris cover of the connected sum

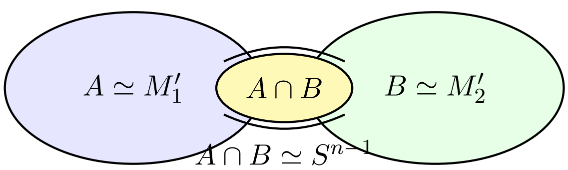


Figure 4: Mayer–Vietoris cover: $A \simeq M_1'$, $B \simeq M_2'$, and $A \cap B \simeq S^{n-1}$.

5.5 Chart 1: Cohomology of the connected sum by degree

Degree	Group for $M_1 \# M_2$	Reason
0	$H^0(M_1 \# M_2) \cong \mathbb{Z}$	$M_1 \# M_2$ is connected
$0 < i < n$	$H^i(M_1 \# M_2) \cong H^i(M_1) \oplus H^i(M_2)$	Mayer–Vietoris and $A \cap B \simeq S^{n-1}$
n	$H^n(M_1 \# M_2) \cong \mathbb{Z}$	$M_1 \# M_2$ is closed and oriented

5.6 Diagram 5: Why degrees 0 and n are not additive

Degree	Direct sum prediction	Actual group
0	$H^0(M_1) \oplus H^0(M_2) \cong \mathbb{Z} \oplus \mathbb{Z}$	$H^0(M_1 \# M_2) \cong \mathbb{Z}$
n	$H^n(M_1) \oplus H^n(M_2) \cong \mathbb{Z} \oplus \mathbb{Z}$	$H^n(M_1 \# M_2) \cong \mathbb{Z}$

The connected sum has one connected component and one fundamental class.

5.7 Diagram 6: Cup product pairing becomes block diagonal

For cohomology classes in middle degrees,

$$H^i(M_1 \# M_2) \cong H^i(M_1) \oplus H^i(M_2).$$

Thus a class is written as

$$(a_1, a_2),$$

and the cup product pairing satisfies

$$((a_1, a_2), (b_1, b_2))_{\#} = (a_1, b_1)_1 + (a_2, b_2)_2.$$

This can be visualized as a block diagonal form:

$$\boxed{Q_{M_1 \# M_2} = \begin{pmatrix} Q_{M_1} & 0 \\ 0 & Q_{M_2} \end{pmatrix}}$$

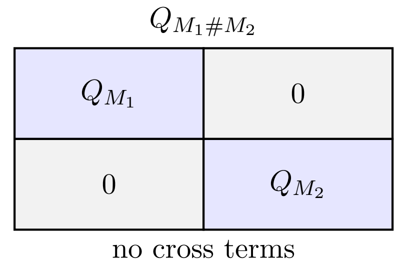


Figure 5: The cup product pairing of a connected sum is the orthogonal direct sum of the pairings of the two summands.

5.8 Diagram 7: Surface connected sums

For closed orientable surfaces,

$$\Sigma_g \# \Sigma_h \cong \Sigma_{g+h}.$$

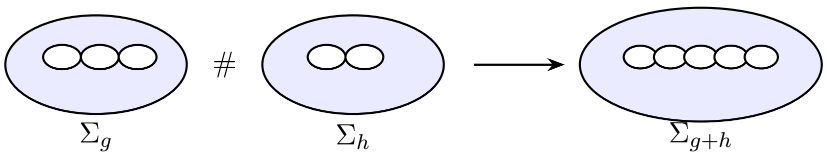


Figure 6: For orientable surfaces, connected sum adds handles: $\Sigma_g \# \Sigma_h \cong \Sigma_{g+h}$.

5.9 Chart 2: Examples and their cohomology

Connected sum	**Resulting space**	**Middle cohomology / pairing**
$S^n \# M$	M	S^n is the identity
$S^n \# S^n$	S^n	degree 0 and n are not additive
$T^2 \# T^2$	Σ_2	$H^1(\Sigma_2) \cong \mathbb{Z}^4$
$\Sigma_g \# \Sigma_h$	Σ_{g+h}	$H^1(\Sigma_{g+h}) \cong \mathbb{Z}^{2(g+h)}$
$\mathbb{CP}^2 \# \mathbb{CP}^2$	4-manifold	$\langle 1 \rangle \oplus \langle 1 \rangle$
$\mathbb{CP}^2 \# \overline{\mathbb{CP}}^2$	4-manifold	$\langle 1 \rangle \oplus \langle -1 \rangle$
$(S^2 \times S^2) \# \mathbb{CP}^2$	4-manifold	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \oplus \langle 1 \rangle$

5.10 Diagram 8: Intersection forms in dimension four

For closed oriented 4-manifolds, the middle cup product pairing is the intersection form on H^2.

Manifold	**Intersection form**
$\mathbb{CP}^2$	$\begin{pmatrix} 1 \end{pmatrix}$
$\overline{\mathbb{CP}}^2$	$\begin{pmatrix} -1 \end{pmatrix}$
$S^2 \times S^2$	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$
$\mathbb{CP}^2 \# \mathbb{CP}^2$	$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$
$\mathbb{CP}^2 \# \overline{\mathbb{CP}}^2$	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$
$(S^2 \times S^2) \# \mathbb{CP}^2$	$\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

5.11 Final visual summary

remove two disks

$\Downarrow$

glue boundary spheres with opposite orientations

$\Downarrow$

$M_1 \# M_2$ is a closed oriented manifold

$\Downarrow$

$$H^i(M_1 \# M_2) \cong H^i(M_1) \oplus H^i(M_2), \quad 0 < i < n$$

$\Downarrow$

$$Q_{M_1 \# M_2} = Q_{M_1} \oplus Q_{M_2}$$

6 Problems 3–5: Intersection Forms and Middle-Dimensional Pairings

Throughout this section, manifolds are assumed to be closed, connected, and $\mathbb{Z}$-orientable, so that Poincaré duality applies.

7 Problem 3

Problem. Suppose M is a $\mathbb{Z}$-orientable 4-manifold. If $H_1(M)$ is free over $\mathbb{Z}$, show that $H_*(M)$ and $H^*(M)$ are free over $\mathbb{Z}$. Assume that this is the case, and choose a basis

$$\langle a_1, \dots, a_n \rangle$$

for $H^2(M)$. Let A be the matrix whose ij-th entry is

$$(a_i, a_j).$$

Show that A is symmetric, and that

$$\det A = \pm 1.$$

Solution.

Since M is a closed orientable 4-manifold, Poincaré duality gives

$$H^k(M) \cong H_{4-k}(M).$$

We also use the universal coefficient theorem for cohomology:

$$0 \longrightarrow \operatorname{Ext}(H_{k-1}(M), \mathbb{Z}) \longrightarrow H^k(M) \longrightarrow \operatorname{Hom}(H_k(M), \mathbb{Z}) \longrightarrow 0.$$

Because $H_1(M)$ is free, we have

$$\operatorname{Ext}(H_1(M), \mathbb{Z}) = 0.$$

Therefore, for $k = 2$, the universal coefficient theorem gives

$$H^2(M) \cong \operatorname{Hom}(H_2(M), \mathbb{Z}).$$

On the other hand, Poincaré duality gives

$$H^2(M) \cong H_2(M).$$

Combining the two isomorphisms, we get

$$H_2(M) \cong \operatorname{Hom}(H_2(M), \mathbb{Z}).$$

Since $H_2(M)$ is finitely generated, this implies that $H_2(M)$ is free. Indeed, if $H_2(M)$ had a nonzero torsion subgroup, then $\mathrm{Hom}(H_2(M), \mathbb{Z})$ would kill that torsion, so $H_2(M)$ could not be isomorphic to its dual.

Now

$$H_0(M) \cong \mathbb{Z}, \qquad H_4(M) \cong \mathbb{Z}.$$

By assumption, $H_1(M)$ is free. Also,

$$H_3(M) \cong H^1(M)$$

by Poincaré duality. The universal coefficient theorem gives

$$H^1(M) \cong \mathrm{Hom}(H_1(M), \mathbb{Z}),$$

because $\mathrm{Ext}(H_0(M), \mathbb{Z}) = 0$. Since $H_1(M)$ is free, $\mathrm{Hom}(H_1(M), \mathbb{Z})$ is also free. Hence $H_3(M)$ is free.

Thus all homology groups $H_*(M)$ are free over $\mathbb{Z}$. Since the homology groups are free, the universal coefficient theorem gives

$$H^k(M) \cong \mathrm{Hom}(H_k(M), \mathbb{Z}),$$

so all cohomology groups $H^*(M)$ are also free over $\mathbb{Z}$.

Now define the intersection pairing on $H^2(M)$ by

$$(a, b) = \langle a \smile b, [M] \rangle.$$

Let $a, b \in H^2(M)$. Since cup product is graded-commutative,

$$a \smile b = (-1)^{2\cdot 2} b \smile a = b \smile a.$$

Therefore

$$(a, b) = \langle a \smile b, [M] \rangle = \langle b \smile a, [M] \rangle = (b, a).$$

Hence the intersection pairing is symmetric, and so the matrix A is symmetric.

Finally, by Poincaré duality, the map

$$H^2(M) \longrightarrow \mathrm{Hom}(H^2(M), \mathbb{Z}), \qquad a \longmapsto \big(b \mapsto (a, b)\big),$$

is an isomorphism. With respect to the basis

$$\langle a_1, \ldots, a_n \rangle$$

and its dual basis, this map is represented by the matrix A. Therefore A is an invertible integer matrix whose inverse is also an integer matrix. Hence A is unimodular, and so

$$\det A = \pm 1.$$

Thus A is symmetric and unimodular.

$$\boxed{A^T = A, \qquad \det A = \pm 1.}$$

8 Problem 4

Problem. Orient $\mathbb{C}P^2$ over $\mathbb{Z}$ so that

$$(a, a) = 1,$$

where

$$\langle a \rangle = H^2(\mathbb{C}P^2),$$

and let $\overline{\mathbb{C}P}^2$ denote the same manifold $\mathbb{C}P^2$ with the opposite orientation. Define

$$X_1 = \mathbb{C}P^2 \# \mathbb{C}P^2, \qquad X_2 = \mathbb{C}P^2 \# \overline{\mathbb{C}P}^2, \qquad X_3 = S^2 \times S^2.$$

Show that

$$H_*(X_1) \cong H_*(X_2) \cong H_*(X_3),$$

but that no two of the X_i are homotopy equivalent.

Solution.

First compute the homology groups. We know that

$$H_k(\mathbb{C}P^2) \cong \begin{cases} \mathbb{Z}, & k = 0, 2, 4, \\ 0, & \text{otherwise.} \end{cases}$$

The connected sum of two closed connected orientable 4-manifolds has top and bottom homology

$$H_0 \cong \mathbb{Z}, \qquad H_4 \cong \mathbb{Z},$$

and in dimension 2, the groups add. Hence

$$H_2(X_1) \cong H_2(X_2) \cong \mathbb{Z}^2.$$

Also,

$$H_1(X_1) = H_1(X_2) = 0, \qquad H_3(X_1) = H_3(X_2) = 0.$$

For $S^2 \times S^2$, by the Künneth formula,

$$H_k(S^2 \times S^2) \cong \begin{cases} \mathbb{Z}, & k = 0, 4, \\ \mathbb{Z}^2, & k = 2, \\ 0, & \text{otherwise.} \end{cases}$$

Therefore

$$H_k(X_i) \cong \begin{cases} \mathbb{Z}, & k = 0, 4, \\ \mathbb{Z}^2, & k = 2, \\ 0, & k = 1, 3, \end{cases}$$

for each $i = 1, 2, 3$. Hence

$$H_*(X_1) \cong H_*(X_2) \cong H_*(X_3).$$

Now we distinguish the spaces using intersection forms.

For $\mathbb{C}P^2$, choose the generator $a \in H^2(\mathbb{C}P^2)$ such that

$$(a, a) = 1.$$

Reversing orientation changes the sign of the fundamental class. Hence the intersection form on $\overline{\mathbb{C}P}^2$ is

$$(a, a) = -1.$$

Therefore, with respect to natural bases,

$$Q_{X_1} \cong \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},$$

and

$$Q_{X_2} \cong \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

For $X_3 = S^2 \times S^2$, let

$$u = [S^2 \times \{\mathrm{pt}\}], \qquad v = [\{\mathrm{pt}\} \times S^2].$$

Then

$$u \cdot u = 0, \qquad v \cdot v = 0, \qquad u \cdot v = 1.$$

Thus

$$Q_{X_3} \cong \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

So the three intersection forms are

$$Q_{X_1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad Q_{X_2} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad Q_{X_3} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

A homotopy equivalence between closed oriented 4-manifolds preserves the intersection form up to sign. Therefore, if two of the X_i were homotopy equivalent, their intersection forms would be isomorphic up to multiplication by -1.

Now Q_{X_1} is definite. Indeed,

$$Q_{X_1}(x, y) = x^2 + y^2,$$

so it is positive definite. Its signature is

$$\sigma(X_1) = 2.$$

On the other hand, Q_{X_2} is indefinite:

$$Q_{X_2}(x, y) = x^2 - y^2,$$

and Q_{X_3} is also indefinite:

$$Q_{X_3}(x, y) = 2xy.$$

Both have signature 0. Therefore X_1 cannot be homotopy equivalent to X_2 or X_3.

It remains to distinguish X_2 and X_3. The form Q_{X_2} is odd because there exists a class with odd self-intersection. For example,

$$(1, 0) \cdot (1, 0) = 1.$$

However, Q_{X_3} is even. Indeed, for any class

$$x = au + bv,$$

we have

$$x \cdot x = (au + bv) \cdot (au + bv) = 2ab,$$

which is always even.

Parity of an integral intersection form is preserved under isomorphism, and it is also preserved after multiplying the form by -1. Hence

$$Q_{X_2} \not\cong Q_{X_3}$$

even up to sign. Therefore X_2 and X_3 are not homotopy equivalent.

Thus the three manifolds have isomorphic homology groups, but no two of them are homotopy equivalent.

$$\boxed{H_*(X_1) \cong H_*(X_2) \cong H_*(X_3), \quad \text{but } X_i \not\simeq X_j \text{ for } i \neq j.}$$

9 Problem 5

Problem. If M is a $\mathbb{Z}$-orientable manifold of dimension $4n + 2$, show that the dimension of

$$H_{2n+1}(M; \mathbb{Q})$$

is even.

Solution.

Assume

$$\dim M = 4n + 2.$$

The middle dimension is therefore

$$2n + 1.$$

Consider the middle-dimensional cup-product pairing over $\mathbb{Q}$:

$$H^{2n+1}(M; \mathbb{Q}) \times H^{2n+1}(M; \mathbb{Q}) \longrightarrow \mathbb{Q},$$

defined by

$$(a, b) = \langle a \smile b, [M] \rangle.$$

By Poincaré duality, this pairing is nondegenerate.

Now let

$$a, b \in H^{2n+1}(M; \mathbb{Q}).$$

Since $2n+1$ is odd, graded commutativity gives

$$a \smile b = (-1)^{(2n+1)(2n+1)} b \smile a.$$

But

$$(2n+1)(2n+1)$$

is odd, so

$$(-1)^{(2n+1)(2n+1)} = -1.$$

Therefore

$$a \smile b = -b \smile a.$$

Hence

$$(a, b) = -(b, a).$$

Thus the middle-dimensional intersection pairing is skew-symmetric.

So $H^{2n+1}(M; \mathbb{Q})$ is a finite-dimensional vector space over $\mathbb{Q}$ equipped with a nondegenerate skew-symmetric bilinear form.

We now use the linear algebra fact that a nondegenerate skew-symmetric bilinear form can exist only on an even-dimensional vector space.

Indeed, suppose the form has matrix A. Since the form is skew-symmetric,

$$A^T = -A.$$

Taking determinants gives

$$\det(A^T) = \det(-A).$$

Since $\det(A^T) = \det(A)$, we get

$$\det(A) = (-1)^m \det(A),$$

where m is the size of the matrix, i.e.

$$m = \dim_{\mathbb{Q}} H^{2n+1}(M; \mathbb{Q}).$$

If m were odd, then

$$(-1)^m = -1,$$

so

$$\det(A) = -\det(A).$$

Therefore

$$2 \det(A) = 0.$$

Over $\mathbb{Q}$, this implies

$$\det(A) = 0,$$

contradicting the nondegeneracy of the pairing. Hence m must be even.

Therefore

$$\dim_{\mathbb{Q}} H^{2n+1}(M;\mathbb{Q})$$

is even.

Finally, over a field, the universal coefficient theorem gives

$$H^{2n+1}(M;\mathbb{Q}) \cong \operatorname{Hom}_{\mathbb{Q}}(H_{2n+1}(M;\mathbb{Q}),\mathbb{Q}).$$

Thus

$$\dim_{\mathbb{Q}} H^{2n+1}(M;\mathbb{Q}) = \dim_{\mathbb{Q}} H_{2n+1}(M;\mathbb{Q}).$$

Hence

$$\boxed{\dim_{\mathbb{Q}} H_{2n+1}(M;\mathbb{Q}) \text{ is even.}}$$

10 Theory for Problems 3–5: Intersection Forms and Middle-Dimensional Pairings

Throughout this section, manifolds are assumed to be closed, connected, and $\mathbb{Z}$-orientable, unless stated otherwise. These assumptions ensure that Poincaré duality applies.

The central theme behind Problems 3–5 is the passage

$$\text{topology of a manifold} \quad\longrightarrow\quad \text{homology and cohomology} \quad\longrightarrow\quad \text{bilinear pairings.}$$

For 4-manifolds, the most important such pairing is the intersection form on $H^2(M;\mathbb{Z})$. For manifolds of dimension $4n+2$, the corresponding middle-dimensional pairing is skew-symmetric.

10.1 Poincaré Duality

Let M be a closed, connected, orientable m-manifold. Then M has a fundamental class

$$[M] \in H_m(M;\mathbb{Z}).$$

Poincaré duality says that cap product with $[M]$ gives isomorphisms

$$H^k(M;\mathbb{Z}) \cong H_{m-k}(M;\mathbb{Z}).$$

For a closed orientable 4-manifold, this gives

$$\begin{aligned} H^0(M) &\cong H_4(M), \\ H^1(M) &\cong H_3(M), \\ H^2(M) &\cong H_2(M), \\ H^3(M) &\cong H_1(M), \end{aligned}$$

and

$$H^4(M) \cong H_0(M).$$

The middle dimension is especially important:

$$H^2(M) \cong H_2(M).$$

This is the reason that $H^2(M)$ carries a natural self-pairing.

10.2 The Universal Coefficient Theorem

The universal coefficient theorem for cohomology describes $H^k(M;\mathbb{Z})$ in terms of the homology groups of M. It gives a short exact sequence

$$0 \longrightarrow \mathrm{Ext}(H_{k-1}(M),\mathbb{Z}) \longrightarrow H^k(M;\mathbb{Z}) \longrightarrow \mathrm{Hom}(H_k(M),\mathbb{Z}) \longrightarrow 0.$$

The important point is that torsion in $H_{k-1}(M)$ contributes to $H^k(M;\mathbb{Z})$ through the Ext-term.

If $H_{k-1}(M)$ is free, then

$$\mathrm{Ext}(H_{k-1}(M),\mathbb{Z}) = 0.$$

Hence

$$H^k(M;\mathbb{Z}) \cong \mathrm{Hom}(H_k(M),\mathbb{Z}).$$

In Problem 3, the assumption

$$H_1(M) \text{ is free over } \mathbb{Z}$$

implies

$$\mathrm{Ext}(H_1(M),\mathbb{Z}) = 0.$$

Therefore

$$H^2(M;\mathbb{Z}) \cong \mathrm{Hom}(H_2(M),\mathbb{Z}).$$

Together with Poincaré duality,

$$H^2(M;\mathbb{Z}) \cong H_2(M;\mathbb{Z}),$$

we obtain

$$H_2(M;\mathbb{Z}) \cong \mathrm{Hom}(H_2(M),\mathbb{Z}).$$

This forces $H_2(M)$ to be free.

10.3 Why Torsion Disappears in the Dual

Let A be a finitely generated abelian group. Then

$$A \cong \mathbb{Z}^r \oplus T,$$

where T is the torsion subgroup.

Then

$$\mathrm{Hom}(A,\mathbb{Z}) \cong \mathrm{Hom}(\mathbb{Z}^r,\mathbb{Z}) \oplus \mathrm{Hom}(T,\mathbb{Z}).$$

But

$$\mathrm{Hom}(T,\mathbb{Z}) = 0,$$

because every homomorphism from a finite torsion group to $\mathbb{Z}$ is zero. Indeed, if $x \in T$, then there exists $m > 0$ such that

$$mx = 0.$$

For a homomorphism $f : T \to \mathbb{Z}$, this implies

$$mf(x) = f(mx) = f(0) = 0.$$

Since $\mathbb{Z}$ has no torsion, we get

$$f(x) = 0.$$

Therefore

$$\mathrm{Hom}(A, \mathbb{Z}) \cong \mathbb{Z}^r.$$

The dual group remembers only the free part of A, not its torsion.

Hence, if a finitely generated abelian group A satisfies

$$A \cong \mathrm{Hom}(A, \mathbb{Z}),$$

then A cannot have torsion. Therefore A must be free.

This is the mechanism used in Problem 3 to prove that $H_2(M)$ is free.

10.4 The Intersection Pairing on a 4-Manifold

Let M be a closed orientable 4-manifold. For classes

$$a, b \in H^2(M; \mathbb{Z}),$$

define

$$(a, b) = \langle a \smile b, [M] \rangle.$$

Here

$$a \smile b \in H^4(M; \mathbb{Z}),$$

and

$$[M] \in H_4(M; \mathbb{Z}).$$

Thus the cup product followed by evaluation on the fundamental class gives an integer:

$$H^2(M; \mathbb{Z}) \times H^2(M; \mathbb{Z}) \longrightarrow \mathbb{Z}.$$

This pairing is called the *intersection form* of M.

Geometrically, if a and b are Poincaré dual to embedded oriented surfaces A and B in M, then

$$(a, b)$$

counts the signed intersection points of A and B. Thus algebraically,

$$(a, b) = \langle a \smile b, [M] \rangle,$$

while geometrically,

$$(a, b) = A \cdot B.$$

10.5 Symmetry of the Intersection Form in Dimension 4

The cup product is graded-commutative:

$$a \smile b = (-1)^{\deg(a)\deg(b)} b \smile a.$$

For $a, b \in H^2(M; \mathbb{Z})$, we have

$$\deg(a) = \deg(b) = 2.$$

Therefore

$$(-1)^{\deg(a)\deg(b)} = (-1)^{2\cdot 2} = (-1)^4 = 1.$$

Hence

$$a \smile b = b \smile a.$$

Therefore

$$(a, b) = \langle a \smile b, [M] \rangle = \langle b \smile a, [M] \rangle = (b, a).$$

So the intersection form of a closed orientable 4-manifold is symmetric. If

$$\langle a_1, \ldots, a_n \rangle$$

is a basis for $H^2(M; \mathbb{Z})$, and if A is the matrix with entries

$$A_{ij} = (a_i, a_j),$$

then symmetry gives

$$A_{ij} = A_{ji}.$$

Equivalently,

$$A^T = A.$$

10.6 Unimodularity of the Intersection Form

Let $L \cong \mathbb{Z}^n$ be a free abelian group, and let

$$Q : L \times L \to \mathbb{Z}$$

be a bilinear form. The form Q is called *unimodular* if the associated map

$$L \longrightarrow \mathrm{Hom}(L, \mathbb{Z}), \qquad x \longmapsto \big(y \mapsto Q(x, y)\big),$$

is an isomorphism.

If Q has matrix A with respect to a basis of L, then this condition is equivalent to saying that A is invertible over $\mathbb{Z}$. Equivalently,

$$\det A = \pm 1.$$

For a closed orientable 4-manifold with free $H^2(M; \mathbb{Z})$, Poincaré duality implies that the map

$$H^2(M; \mathbb{Z}) \longrightarrow \mathrm{Hom}(H^2(M; \mathbb{Z}), \mathbb{Z}), \qquad a \longmapsto \big(b \mapsto (a, b)\big),$$

is an isomorphism.

Therefore the intersection matrix is unimodular:

$$\det A = \pm 1.$$

Thus, in Problem 3, the intersection matrix A is both symmetric and unimodular:

$$A^T = A, \qquad \det A = \pm 1.$$

10.7 Intersection Forms as Homotopy Invariants

For closed oriented 4-manifolds, the intersection form is a homotopy invariant up to sign.

Let

$$f : M \to N$$

be a homotopy equivalence between closed oriented 4-manifolds. Then f induces an isomorphism

$$f^* : H^2(N; \mathbb{Z}) \to H^2(M; \mathbb{Z}).$$

If f preserves orientation, then

$$f_*[M] = [N],$$

and the intersection forms are preserved. If f reverses orientation, then

$$f_*[M] = -[N],$$

and the intersection form is multiplied by -1.

Therefore, if two closed oriented 4-manifolds are homotopy equivalent, then their intersection forms must be isomorphic up to sign.

This is the main tool used in Problem 4.

10.8 The Basic Intersection Forms in Problem 4

Problem 4 considers the manifolds

$$X_1 = \mathbb{C}P^2 \# \mathbb{C}P^2,$$

$$X_2 = \mathbb{C}P^2 \# \overline{\mathbb{C}P}^2,$$

and

$$X_3 = S^2 \times S^2.$$

Although these manifolds have the same homology groups, their intersection forms are different.

The Form of $\mathbb{C}P^2$

The cohomology ring of $\mathbb{C}P^2$ is
$$H^*(\mathbb{C}P^2;\mathbb{Z}) \cong \mathbb{Z}[a]/(a^3),$$
where
$$\deg(a) = 2.$$
The top-dimensional class is
$$a^2 \in H^4(\mathbb{C}P^2;\mathbb{Z}).$$
Choosing the standard orientation means that
$$\langle a^2, [\mathbb{C}P^2]\rangle = 1.$$
Therefore
$$(a, a) = 1.$$
Thus the intersection form of $\mathbb{C}P^2$ is
$$[1].$$

The Form of $\overline{\mathbb{C}P}^2$

The manifold
$$\overline{\mathbb{C}P}^2$$
is the same underlying smooth manifold as $\mathbb{C}P^2$, but with the opposite orientation.

Changing orientation replaces the fundamental class by its negative:
$$[\overline{\mathbb{C}P}^2] = -[\mathbb{C}P^2].$$
Therefore
$$(a, a)_{\overline{\mathbb{C}P}^2} = \langle a^2, [\overline{\mathbb{C}P}^2]\rangle = -\langle a^2, [\mathbb{C}P^2]\rangle = -1.$$
Thus the intersection form of $\overline{\mathbb{C}P}^2$ is
$$[-1].$$

Connected Sums and Direct Sums of Forms

For closed orientable 4-manifolds, the intersection form of a connected sum is the direct sum of the intersection forms:
$$Q_{M\#N} \cong Q_M \oplus Q_N.$$
Therefore
$$Q_{\mathbb{C}P^2\#\mathbb{C}P^2} = [1] \oplus [1] = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$
Similarly,
$$Q_{\mathbb{C}P^2\#\overline{\mathbb{C}P}^2} = [1] \oplus [-1] = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The Form of $S^2 \times S^2$

For $S^2 \times S^2$, let

$$u = [S^2 \times \{\mathrm{pt}\}], \qquad v = [\{\mathrm{pt}\} \times S^2].$$

The surfaces $S^2 \times \{\mathrm{pt}\}$ and $\{\mathrm{pt}\} \times S^2$ intersect transversely in one point, so

$$u \cdot v = 1.$$

Each has self-intersection zero:

$$u \cdot u = 0, \qquad v \cdot v = 0.$$

Therefore the intersection form of $S^2 \times S^2$ is

$$Q_{S^2 \times S^2} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

This form is called the *hyperbolic form.*

10.9 Definite, Indefinite, Even, and Odd Forms

Intersection forms can be distinguished using algebraic properties such as signature, definiteness, and parity.

Definite Forms

A symmetric bilinear form Q is called *positive definite* if

$$Q(x, x) > 0$$

for every nonzero vector x.

The form

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

is positive definite because

$$Q(x, y) = x^2 + y^2.$$

Therefore

$$\mathbb{C}P^2 \# \mathbb{C}P^2$$

has a positive definite intersection form.

Indefinite Forms

A symmetric bilinear form is called *indefinite* if it takes both positive and negative values.

The form

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

is indefinite because

$$Q(x, y) = x^2 - y^2.$$

The hyperbolic form

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

is also indefinite, because

$$Q(x, y) = 2xy.$$

For example,

$$Q(1, 1) = 2,$$

but

$$Q(1, -1) = -2.$$

Thus

$$\mathbb{C}P^2 \# \overline{\mathbb{C}P}^2$$

and

$$S^2 \times S^2$$

both have indefinite intersection forms.

Signature

The signature of a symmetric bilinear form is

$$\sigma(Q) = b_+(Q) - b_-(Q),$$

where $b_+(Q)$ is the number of positive eigenvalues and $b_-(Q)$ is the number of negative eigenvalues.

For

$$Q_{X_1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},$$

we have

$$b_+ = 2, \qquad b_- = 0.$$

Therefore

$$\sigma(X_1) = 2.$$

For

$$Q_{X_2} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

we have

$$b_+ = 1, \qquad b_- = 1.$$

Therefore

$$\sigma(X_2) = 0.$$

For

$$Q_{X_3} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

the eigenvalues are 1 and -1, so

$$b_+ = 1, \qquad b_- = 1.$$

Therefore

$$\sigma(X_3) = 0.$$

Hence X_1 is distinguished from X_2 and X_3 by signature.

Even and Odd Forms

An integral symmetric form Q is called *even* if

$$Q(x, x) \in 2\mathbb{Z}$$

for every integral class x.

It is called *odd* if there exists an integral class x such that

$$Q(x, x)$$

is odd.

The form

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

is odd because

$$Q(1, 0) = 1.$$

The hyperbolic form

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

is even. Indeed, if

$$x = au + bv,$$

then

$$x \cdot x = (au + bv) \cdot (au + bv) = 2ab.$$

Hence

$$x \cdot x \in 2\mathbb{Z}.$$

Therefore

$$\mathbb{C}P^2 \# \overline{\mathbb{C}P}^2$$

and

$$S^2 \times S^2$$

are not homotopy equivalent.

10.10 Same Homology, Different Homotopy Type

Problem 4 illustrates an important principle:

homology groups alone do not determine homotopy type.

Indeed,

$$H_*(X_1) \cong H_*(X_2) \cong H_*(X_3),$$

but the cup product structures are different. Since the intersection form is constructed from the cup product, it detects information that the homology groups alone do not see.

Thus we have the hierarchy

$$\text{homology groups} \quad < \quad \text{cohomology ring} \quad < \quad \text{homotopy type.}$$

In these examples, the homology groups are the same, but the intersection forms are different. Therefore the homotopy types are different.

10.11 Middle-Dimensional Pairings in General Dimension

Let M be a closed orientable m-manifold, and let F be a field. The cup product and evaluation on the fundamental class give a pairing

$$H^k(M;F) \times H^{m-k}(M;F) \longrightarrow F$$

defined by

$$(a,b) = \langle a \smile b, [M] \rangle.$$

If $m = 2r$ is even and $k = r$, this becomes a pairing on the middle cohomology:

$$H^r(M;F) \times H^r(M;F) \longrightarrow F.$$

The symmetry of this middle-dimensional pairing depends on whether r is even or odd. By graded commutativity,

$$a \smile b = (-1)^{r^2} b \smile a.$$

Therefore:

$$\begin{cases} r \text{ even} & \Longrightarrow \text{the pairing is symmetric,} \\ r \text{ odd} & \Longrightarrow \text{the pairing is skew-symmetric.} \end{cases}$$

10.12 Why Dimension $4n + 2$ Gives a Skew-Symmetric Pairing

Suppose

$$\dim M = 4n + 2.$$

Then the middle dimension is

$$\frac{4n+2}{2} = 2n + 1.$$

This number is odd. Hence for

$$a, b \in H^{2n+1}(M; \mathbb{Q}),$$

graded commutativity gives

$$a \smile b = (-1)^{(2n+1)(2n+1)} b \smile a.$$

Since

$$(2n+1)^2$$

is odd, we get

$$a \smile b = -b \smile a.$$

Therefore the middle-dimensional pairing

$$H^{2n+1}(M; \mathbb{Q}) \times H^{2n+1}(M; \mathbb{Q}) \longrightarrow \mathbb{Q}$$

is skew-symmetric:

$$(a, b) = -(b, a).$$

By Poincaré duality, this pairing is also nondegenerate.

Thus

$$H^{2n+1}(M; \mathbb{Q})$$

is a finite-dimensional vector space over $\mathbb{Q}$ equipped with a nondegenerate skew-symmetric bilinear form.

10.13 Linear Algebra Fact: Skew-Symmetric Forms Have Even Dimension

Let V be a finite-dimensional vector space over a field of characteristic not equal to 2, for example $\mathbb{Q}$. A bilinear form

$$\omega : V \times V \to \mathbb{Q}$$

is skew-symmetric if

$$\omega(x, y) = -\omega(y, x).$$

If ω is nondegenerate, then $\dim V$ must be even.

To see this, let A be the matrix of ω. Since ω is skew-symmetric,

$$A^T = -A.$$

Taking determinants gives

$$\det(A^T) = \det(-A).$$

Since

$$\det(A^T) = \det(A),$$

and if $m = \dim V$, then

$$\det(-A) = (-1)^m \det(A).$$

Therefore

$$\det(A) = (-1)^m \det(A).$$

If m were odd, then

$$(-1)^m = -1,$$

so

$$\det(A) = -\det(A).$$

Hence

$$2\det(A) = 0.$$

Over $\mathbb{Q}$, this implies

$$\det(A) = 0.$$

But $\det(A) = 0$ contradicts nondegeneracy. Therefore m must be even.

10.14 Application to Problem 5

Let M be a closed orientable manifold of dimension

$$4n + 2.$$

Then the middle-dimensional rational cohomology group

$$H^{2n+1}(M; \mathbb{Q})$$

has a nondegenerate skew-symmetric pairing.

Therefore

$$\dim_{\mathbb{Q}} H^{2n+1}(M; \mathbb{Q})$$

is even.

Over a field, the universal coefficient theorem gives

$$H^{2n+1}(M; \mathbb{Q}) \cong \mathrm{Hom}_{\mathbb{Q}}\Big(H_{2n+1}(M; \mathbb{Q}), \mathbb{Q}\Big).$$

Hence

$$\dim_{\mathbb{Q}} H^{2n+1}(M; \mathbb{Q}) = \dim_{\mathbb{Q}} H_{2n+1}(M; \mathbb{Q}).$$

Therefore

$$\dim_{\mathbb{Q}} H_{2n+1}(M; \mathbb{Q})$$

is even.

Equivalently, the middle Betti number

$$b_{2n+1}(M) = \dim_{\mathbb{Q}} H_{2n+1}(M; \mathbb{Q})$$

is even.

10.15 Examples

Example 1: Closed Orientable Surfaces

Let $M = \Sigma_g$ be a closed orientable surface of genus g. Then

$$\dim M = 2 = 4 \cdot 0 + 2.$$

Thus Problem 5 applies with

$$2n + 1 = 1.$$

We know that

$$H_1(\Sigma_g; \mathbb{Q}) \cong \mathbb{Q}^{2g}.$$

Therefore

$$\dim_{\mathbb{Q}} H_1(\Sigma_g; \mathbb{Q}) = 2g,$$

which is even.

The intersection pairing on $H_1(\Sigma_g; \mathbb{Q})$ is the classical symplectic form.

Example 2: The Torus T^2

For

$$T^2 = S^1 \times S^1,$$

we have

$$H_1(T^2; \mathbb{Q}) \cong \mathbb{Q}^2.$$

With respect to the standard basis given by the two circle factors, the intersection pairing has matrix

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

This matrix is skew-symmetric and nondegenerate. Hence

$$b_1(T^2) = 2.$$

Example 3: A 6-Manifold

Let

$$\dim M = 6.$$

Then

$$6 = 4 \cdot 1 + 2.$$

So Problem 5 applies with

$$2n + 1 = 3.$$

Therefore

$$\dim_{\mathbb{Q}} H_3(M; \mathbb{Q})$$

must be even.

For example, let
$$M = S^3 \times S^3.$$
By the Künneth formula,
$$H_3(S^3 \times S^3; \mathbb{Q}) \cong \mathbb{Q}^2.$$
Thus
$$\dim_{\mathbb{Q}} H_3(S^3 \times S^3; \mathbb{Q}) = 2,$$
which is even.

10.16 Summary Table

The contrast between Problems 3–4 and Problem 5 is summarized as follows:

$\dim M$	middle degree	type of pairing	main consequence
4	2	symmetric	intersection form on $H^2(M)$
$4n+2$	$2n+1$	skew-symmetric	middle Betti number is even

Thus:
$$\text{Poincaré duality} + \text{cup product} = \text{intersection pairing}.$$

For 4-manifolds, the middle degree is even, so the intersection form is symmetric:
$$(a, b) = (b, a).$$

For manifolds of dimension $4n+2$, the middle degree is odd, so the middle-dimensional pairing is skew-symmetric:
$$(a, b) = -(b, a).$$

Consequently:
$$\boxed{\text{In dimension 4, intersection forms distinguish many manifolds.}}$$
and
$$\boxed{\text{In dimension } 4n+2, \text{ the middle rational Betti number is even.}}$$

10.17 Diagrams and Charts for Problems 3–5

Diagram 1: From Poincaré Duality to the Intersection Form

The following diagram summarizes the logical structure behind Problem 3. The intersection form appears because $H^2(M)$ is the middle cohomology group of a closed orientable 4-manifold.

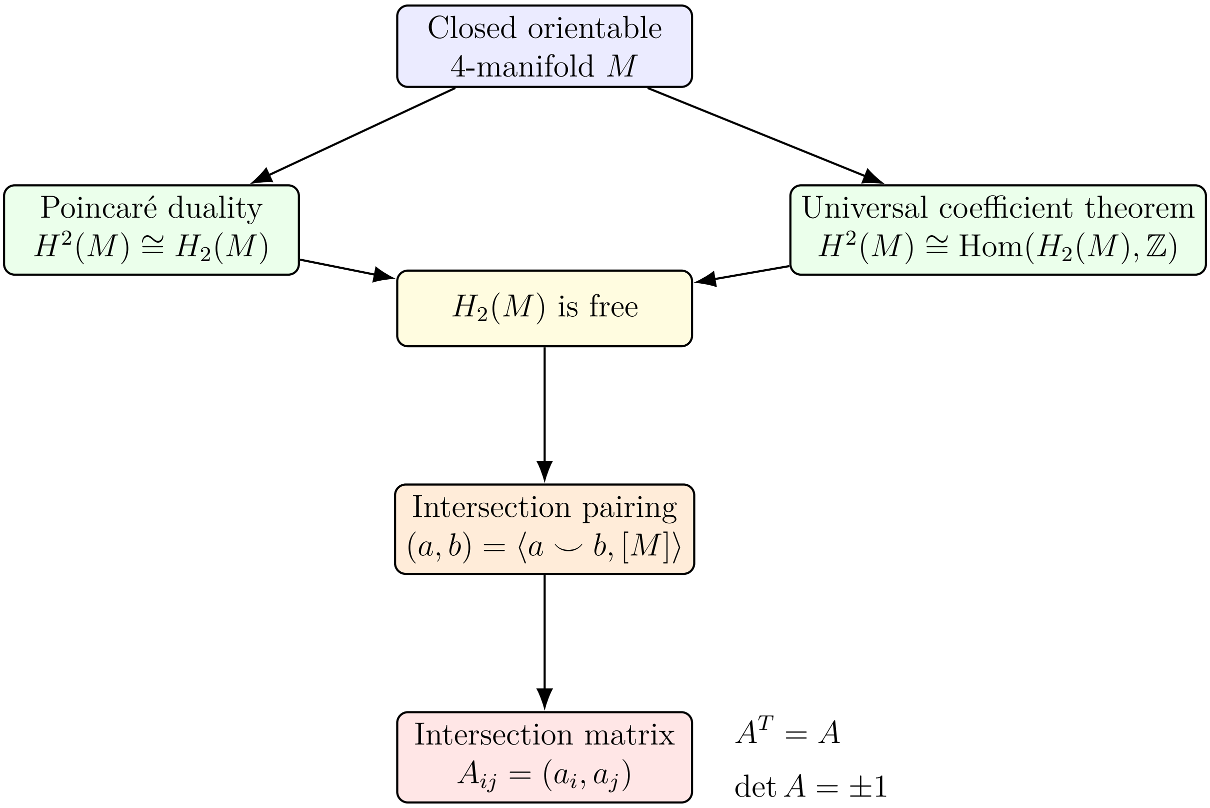


Diagram 2: The Middle-Dimensional Pairing

For a closed orientable m-manifold, the cup product and evaluation on the fundamental class give the pairing

$$H^k(M) \times H^{m-k}(M) \longrightarrow \mathbb{Z}.$$

In the middle dimension, this becomes a self-pairing.

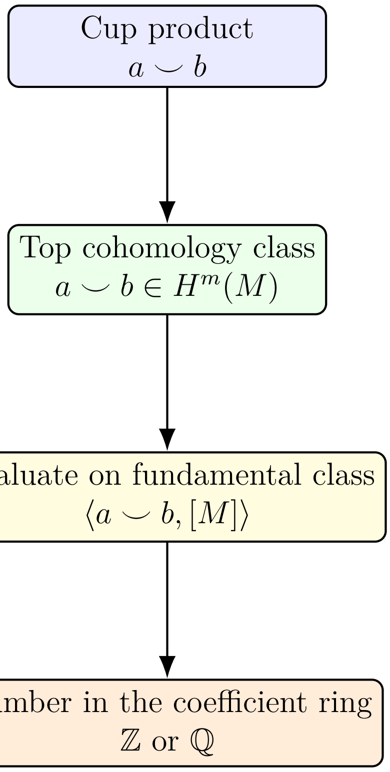


Chart 1: Symmetric versus Skew-Symmetric Middle Pairings

The type of the middle-dimensional pairing depends on the parity of the middle degree.

Dimension of M	Middle degree	Sign from graded commutativity	Type of pairing
4	2	$(-1)^{2\cdot 2} = +1$	symmetric
$4n$	$2n$	$(-1)^{(2n)^2} = +1$	symmetric
$4n+2$	$2n+1$	$(-1)^{(2n+1)^2} = -1$	skew-symmetric

Thus:

$$\dim M = 4 \quad \Longrightarrow \quad (a, b) = (b, a),$$

while

$$\dim M = 4n+2 \quad \Longrightarrow \quad (a, b) = -(b, a).$$

Chart 2: The Three Manifolds in Problem 4

The following table shows why the manifolds in Problem 4 have the same homology but different homotopy types.

Manifold	H_2	Intersection form	Signature	Parity
$X_1 = \mathbb{C}P^2 \# \mathbb{C}P^2$	$\mathbb{Z}^2$	$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$	2	odd
$X_2 = \mathbb{C}P^2 \# \overline{\mathbb{C}P}^2$	$\mathbb{Z}^2$	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$	0	odd
$X_3 = S^2 \times S^2$	$\mathbb{Z}^2$	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$	0	even

Therefore,

$$H_*(X_1) \cong H_*(X_2) \cong H_*(X_3),$$

but their intersection forms are not equivalent. Hence no two of the X_i are homotopy equivalent.

Diagram 3: Intersection Forms of X_1, X_2, and X_3

$$\begin{tikzcd} \begin{array}{c} X_1 = \mathbb{C}P^2 \# \mathbb{C}P^2 \\ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \\ \text{positive definite} \end{array} \arrow[r, "\sigma = 2 \neq 0"] & \begin{array}{c} X_2 = \mathbb{C}P^2 \# \overline{\mathbb{C}P}^2 \\ \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \\ \text{indefinite, odd} \end{array} \arrow[r, "\text{odd} \neq \text{even}"] & \begin{array}{c} X_3 = S^2 \times S^2 \\ \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \\ \text{indefinite, even} \end{array} \end{tikzcd}$$

Chart 3: How the Intersection Forms Distinguish the Manifolds

Pair of manifolds	Distinguishing invariant	Reason
X_1 and X_2	signature	$\sigma(X_1) = 2, \quad \sigma(X_2) = 0$
X_1 and X_3	signature	$\sigma(X_1) = 2, \quad \sigma(X_3) = 0$
X_2 and X_3	parity	Q_{X_2} is odd, but Q_{X_3} is even

Diagram 4: Even-Dimensionality from a Skew-Symmetric Pairing

Problem 5 reduces to the following linear algebra principle.

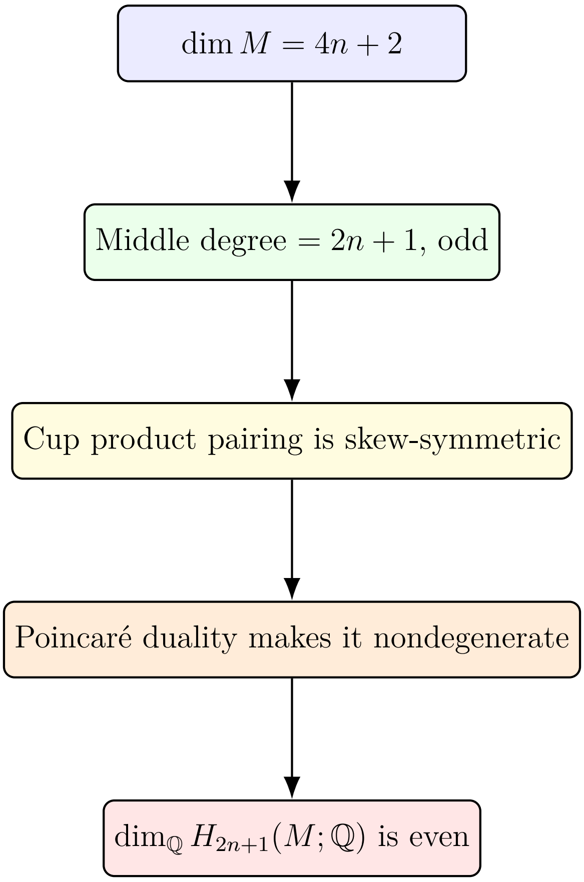


Chart 4: Examples of Problem 5

Manifold	$\dim M$	Middle homology	Middle Betti number
Σ_g	2	$H_1(\Sigma_g;\mathbb{Q}) \cong \mathbb{Q}^{2g}$	$2g$
T^2	2	$H_1(T^2;\mathbb{Q}) \cong \mathbb{Q}^2$	2
$S^3 \times S^3$	6	$H_3(S^3 \times S^3;\mathbb{Q}) \cong \mathbb{Q}^2$	2
$S^{2n+1} \times S^{2n+1}$	$4n+2$	$H_{2n+1} \cong \mathbb{Q}^2$	2

All middle Betti numbers in the table are even, as predicted by Problem 5.

Diagram 5: Standard Skew-Symmetric Form

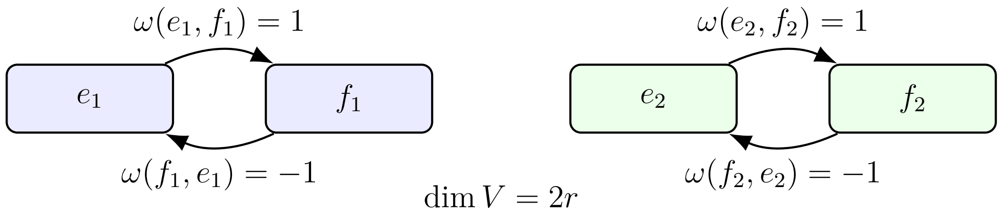
A nondegenerate skew-symmetric form has a standard block form. In a suitable basis, its matrix is made from blocks

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

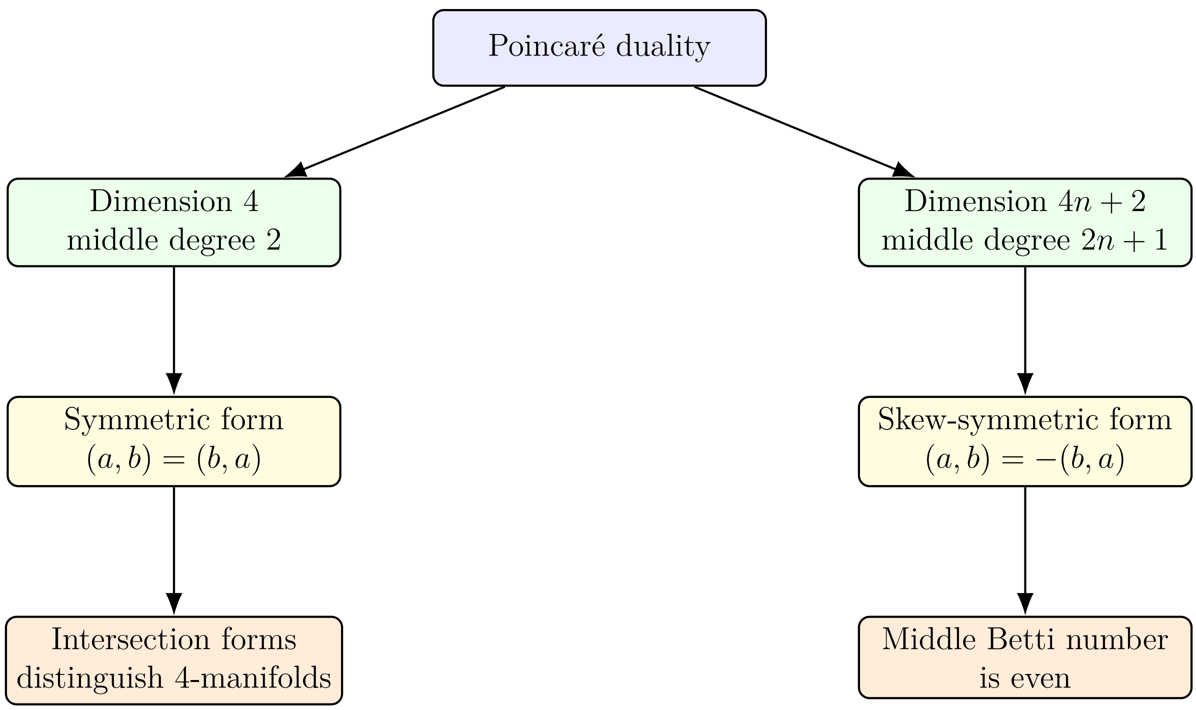
For example, in dimension 4, a standard skew-symmetric matrix is

$$\begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{pmatrix}.$$

This illustrates why the dimension must be even: the basis vectors come in pairs.



Summary Diagram



11 Problems 6–8: Thom Classes, Bézout's Theorem, and Sphere Bundles

Throughout this section, cohomology is taken with integer coefficients unless otherwise stated.

11.1 Problem 6

Problem.

(a) Suppose u_i generates $H^{n_i}(\mathbb{R}^{n_i} \mid 0)$ for $i = 1, 2$. If

$$p_i : \mathbb{R}^{n_1} \times \mathbb{R}^{n_2} \longrightarrow \mathbb{R}^{n_i}$$

is the projection, show that

$$p_1^*(u_1) \smile p_2^*(u_2)$$

generates

$$H^{n_1+n_2}(\mathbb{R}^{n_1} \times \mathbb{R}^{n_2} \mid 0).$$

(b) Now suppose that

$$\pi_i : E_i \to B \qquad (i = 1, 2)$$

are vector bundles over B, and that U_i is a Thom class for E_i. If

$$p_i : E_1 \oplus E_2 \to E_i$$

is the projection, show that

$$p_1^*(U_1) \smile p_2^*(U_2)$$

is a Thom class for $E_1 \oplus E_2$.

(c) Deduce that $E_1 \oplus E_2$ can be oriented so that

$$e(E_1 \oplus E_2) = e(E_1) \smile e(E_2).$$

Solution.

Part (a)

We use the notation

$$H^k(\mathbb{R}^n \mid 0) = H^k(\mathbb{R}^n, \mathbb{R}^n \setminus \{0\}).$$

It is well known that

$$H^k(\mathbb{R}^n, \mathbb{R}^n \setminus \{0\}) \cong \begin{cases} \mathbb{Z}, & k = n, \\ 0, & k \neq n. \end{cases}$$

Hence

$$H^{n_i}(\mathbb{R}^{n_i} \mid 0) \cong \mathbb{Z}$$

for $i = 1, 2$, and

$$H^{n_1+n_2}(\mathbb{R}^{n_1} \times \mathbb{R}^{n_2} \mid 0) \cong \mathbb{Z}.$$

Let

$$u_i \in H^{n_i}(\mathbb{R}^{n_i}, \mathbb{R}^{n_i} \setminus \{0\})$$

be a generator. The product

$$p_1^*(u_1) \smile p_2^*(u_2)$$

is the local cohomology version of the external product

$$u_1 \times u_2.$$

Indeed, pulling back by the two projections and taking the cup product gives the class supported at the common zero

$$(0,0) \in \mathbb{R}^{n_1} \times \mathbb{R}^{n_2}.$$

By the Künneth theorem for these local cohomology groups, the external product induces an isomorphism

$$H^{n_1}(\mathbb{R}^{n_1} \mid 0) \otimes H^{n_2}(\mathbb{R}^{n_2} \mid 0) \cong H^{n_1+n_2}(\mathbb{R}^{n_1} \times \mathbb{R}^{n_2} \mid 0).$$

Since both u_1 and u_2 are generators of infinite cyclic groups, their product is again a generator, up to sign.

Therefore

$$p_1^*(u_1) \smile p_2^*(u_2)$$

generates

$$H^{n_1+n_2}(\mathbb{R}^{n_1} \times \mathbb{R}^{n_2} \mid 0).$$

Thus

$$\boxed{p_1^*(u_1) \smile p_2^*(u_2) \text{ is a generator of } H^{n_1+n_2}(\mathbb{R}^{n_1} \times \mathbb{R}^{n_2} \mid 0).}$$

Part (b)

Let

$$\pi_i : E_i \to B$$

be vector bundles of ranks n_i. A Thom class for E_i is a class

$$U_i \in H^{n_i}(E_i, E_i \setminus B)$$

whose restriction to every fiber

$$(E_i)_b \cong \mathbb{R}^{n_i}$$

is a generator of

$$H^{n_i}((E_i)_b, (E_i)_b \setminus \{0\}) \cong \mathbb{Z}.$$

Now consider the direct sum bundle

$$E_1 \oplus E_2 \to B.$$

Its fiber over $b \in B$ is

$$(E_1 \oplus E_2)_b \cong (E_1)_b \oplus (E_2)_b \cong \mathbb{R}^{n_1} \oplus \mathbb{R}^{n_2}.$$

Let

$$p_i : E_1 \oplus E_2 \to E_i$$

be the bundle projection. Consider the class

$$U = p_1^*(U_1) \smile p_2^*(U_2).$$

This class has degree

$$n_1 + n_2.$$

Moreover, its support is along the zero section of $E_1 \oplus E_2$, so

$$U \in H^{n_1+n_2}(E_1 \oplus E_2, (E_1 \oplus E_2) \setminus B).$$

We must show that U restricts to a generator on each fiber.

Fix $b \in B$. Restricting U to the fiber gives

$$U|_{(E_1 \oplus E_2)_b} = p_1^*(U_1|_{(E_1)_b}) \smile p_2^*(U_2|_{(E_2)_b}).$$

Since U_i is a Thom class, each restriction

$$U_i|_{(E_i)_b}$$

is a generator of the local cohomology group of the fiber:

$$H^{n_i}((E_i)_b, (E_i)_b \setminus \{0\}).$$

By part (a), the cup product of these two pulled-back generators is a generator of

$$H^{n_1+n_2}\Big((E_1)_b \oplus (E_2)_b, ((E_1)_b \oplus (E_2)_b) \setminus \{0\}\Big).$$

Therefore the restriction of

$$p_1^*(U_1) \smile p_2^*(U_2)$$

to every fiber is a generator. Hence this class is a Thom class for $E_1 \oplus E_2$.

Thus

$$\boxed{p_1^*(U_1) \smile p_2^*(U_2) \text{ is a Thom class for } E_1 \oplus E_2.}$$

Part (c)

The choice of a Thom class determines an orientation of a vector bundle. By part (b), the class

$$U = p_1^*(U_1) \smile p_2^*(U_2)$$

is a Thom class for $E_1 \oplus E_2$. We orient $E_1 \oplus E_2$ using this Thom class.

Let

$$s : B \to E_1 \oplus E_2$$

be the zero section of $E_1 \oplus E_2$. By definition, the Euler class of an oriented vector bundle is the pullback of its Thom class along the zero section. Therefore

$$e(E_1 \oplus E_2) = s^*(U).$$

Using the formula for U, we get

$$e(E_1 \oplus E_2) = s^*\Big(p_1^*(U_1) \smile p_2^*(U_2)\Big).$$

By naturality of cup products,

$$e(E_1 \oplus E_2) = s^*p_1^*(U_1) \smile s^*p_2^*(U_2).$$

But

$$p_i \circ s : B \to E_i$$

is precisely the zero section of E_i. Hence

$$s^*p_i^*(U_i) = (p_i \circ s)^*(U_i) = e(E_i).$$

Therefore

$$e(E_1 \oplus E_2) = e(E_1) \smile e(E_2).$$

Thus

$$\boxed{e(E_1 \oplus E_2) = e(E_1) \smile e(E_2).}$$

11.2 Problem 7

Problem. If

$$p \in \mathbb{C}[z_0, z_1, z_2]$$

is a homogeneous polynomial, define

$$V_p = \{[z_0 : z_1 : z_2] \in \mathbb{CP}^2 \mid p(z_0, z_1, z_2) = 0\}.$$

If p and q are chosen so that V_p and V_q are embedded submanifolds of $\mathbb{CP}^2$ which intersect transversely, show that V_p and V_q intersect in precisely

$$(\deg p)(\deg q)$$

points.

Solution.

Let

$$d = \deg p, \qquad e = \deg q.$$

The zero set V_p is a complex hypersurface in $\mathbb{CP}^2$, hence a complex curve. Since V_p is assumed to be an embedded submanifold, it defines a homology class

$$[V_p] \in H_2(\mathbb{CP}^2; \mathbb{Z}).$$

Let

$$a \in H^2(\mathbb{CP}^2; \mathbb{Z})$$

be the standard generator. Then

$$H^*(\mathbb{CP}^2; \mathbb{Z}) \cong \mathbb{Z}[a]/(a^3), \qquad \deg a = 2,$$

and

$$\langle a^2, [\mathbb{CP}^2]\rangle = 1.$$

A homogeneous polynomial of degree d defines a section of the complex line bundle

$$\mathcal{O}(d) \to \mathbb{CP}^2.$$

Its zero set V_p has Poincaré dual equal to the first Chern class of $\mathcal{O}(d)$:

$$[V_p]^\vee = c_1(\mathcal{O}(d)).$$

Since

$$c_1(\mathcal{O}(d)) = d\,a,$$

we obtain

$$[V_p]^\vee = d\,a.$$

Similarly,

$$[V_q]^\vee = e\,a.$$

The algebraic intersection number of V_p and V_q is therefore

$$V_p \cdot V_q = \langle [V_p]^\vee \smile [V_q]^\vee, [\mathbb{CP}^2]\rangle.$$

Substituting the Poincaré dual classes gives

$$V_p \cdot V_q = \langle (d\,a) \smile (e\,a), [\mathbb{CP}^2]\rangle.$$

Hence

$$V_p \cdot V_q = de\langle a^2, [\mathbb{CP}^2]\rangle.$$

Since

$$\langle a^2, [\mathbb{CP}^2]\rangle = 1,$$

we get

$$V_p \cdot V_q = de.$$

Because V_p and V_q are complex submanifolds of the complex surface $\mathbb{CP}^2$, every transverse intersection point has local intersection number $+1$. Therefore the algebraic intersection number is equal to the actual number of intersection points.

Thus

$$|V_p \cap V_q| = de = (\deg p)(\deg q).$$

Therefore

$$\boxed{|V_p \cap V_q| = (\deg p)(\deg q).}$$

This is the smooth transverse case of Bézout's theorem for projective plane curves.

11.3 Problem 8

Problem. Let E be the tangent bundle of $\mathbb{CP}^n$. Compute

$$H_*(S(E)).$$

Solution.

Let

$$B = \mathbb{CP}^n, \qquad E = T\mathbb{CP}^n.$$

Since $\mathbb{CP}^n$ is a complex manifold of complex dimension n, its tangent bundle $T\mathbb{CP}^n$ is a complex vector bundle of complex rank n. As a real vector bundle, it has rank

$$2n.$$

Therefore the sphere bundle

$$S(E) = S(T\mathbb{CP}^n)$$

has fiber

$$S^{2n-1}.$$

So we have an oriented sphere bundle

$$S^{2n-1} \longrightarrow S(T\mathbb{CP}^n) \xrightarrow{\pi} \mathbb{CP}^n.$$

Step 1: Cohomology of $\mathbb{CP}^n$

The cohomology ring of complex projective space is

$$H^*(\mathbb{CP}^n; \mathbb{Z}) \cong \mathbb{Z}[x]/(x^{n+1}), \qquad \deg x = 2.$$

Hence

$$H^{2k}(\mathbb{CP}^n; \mathbb{Z}) \cong \mathbb{Z} \qquad (0 \le k \le n),$$

and

$$H^{2k+1}(\mathbb{CP}^n; \mathbb{Z}) = 0.$$

Step 2: Euler class of $T\mathbb{CP}^n$

For $\mathbb{CP}^n$, the total Chern class of the tangent bundle is

$$c(T\mathbb{CP}^n) = (1+x)^{n+1}.$$

Therefore the top Chern class is

$$c_n(T\mathbb{CP}^n) = (n+1)x^n.$$

Since $T\mathbb{CP}^n$ is a complex vector bundle, the Euler class of its underlying oriented real bundle is the top Chern class:

$$e(T\mathbb{CP}^n) = c_n(T\mathbb{CP}^n).$$

Hence

$$e(E) = e(T\mathbb{CP}^n) = (n+1)x^n.$$

Step 3: The Gysin sequence

For an oriented sphere bundle coming from an oriented real vector bundle of rank $2n$, the Gysin sequence has the form

$$\cdots \longrightarrow H^{q-2n}(\mathbb{CP}^n) \xrightarrow{\smile e(E)} H^q(\mathbb{CP}^n) \longrightarrow H^q(S(E)) \longrightarrow H^{q-2n+1}(\mathbb{CP}^n) \xrightarrow{\smile e(E)} H^{q+1}(\mathbb{CP}^n) \longrightarrow \cdots.$$

Since

$$H^{\text{odd}}(\mathbb{CP}^n) = 0,$$

this sequence simplifies significantly.

Step 4: Degrees below $2n-1$

For

$$q < 2n - 1,$$

we have

$$H^{q-2n}(\mathbb{CP}^n) = 0$$

and

$$H^{q-2n+1}(\mathbb{CP}^n) = 0.$$

Therefore the Gysin sequence gives

$$H^q(S(E)) \cong H^q(\mathbb{CP}^n).$$

Hence

$$H^q(S(E)) \cong \mathbb{Z}$$

for even q satisfying

$$0 \leq q \leq 2n - 2,$$

and

$$H^q(S(E)) = 0$$

for odd $q < 2n - 1$.

Step 5: The middle part

Now take

$$q = 2n - 1$$

and

$$q = 2n.$$

The relevant portion of the Gysin sequence is

$$0 \longrightarrow H^{2n-1}(S(E)) \longrightarrow H^0(\mathbb{CP}^n) \xrightarrow{\smile e(E)} H^{2n}(\mathbb{CP}^n) \longrightarrow H^{2n}(S(E)) \longrightarrow 0.$$

Since

$$H^0(\mathbb{CP}^n) \cong \mathbb{Z}, \qquad H^{2n}(\mathbb{CP}^n) \cong \mathbb{Z},$$

and

$$e(E) = (n+1)x^n,$$

the map

$$H^0(\mathbb{CP}^n) \longrightarrow H^{2n}(\mathbb{CP}^n)$$

is multiplication by $n+1$. Therefore the exact sequence becomes

$$0 \longrightarrow H^{2n-1}(S(E)) \longrightarrow \mathbb{Z} \xrightarrow{\times(n+1)} \mathbb{Z} \longrightarrow H^{2n}(S(E)) \longrightarrow 0.$$

The map

$$\mathbb{Z} \xrightarrow{\times(n+1)} \mathbb{Z}$$

is injective, so

$$H^{2n-1}(S(E)) = 0.$$

Its cokernel is

$$\mathbb{Z}/(n+1).$$

Hence

$$H^{2n}(S(E)) \cong \mathbb{Z}/(n+1).$$

Step 6: Degrees above $2n$

For

$$q > 2n,$$

we have

$$H^q(\mathbb{CP}^n) = 0.$$

The Gysin sequence then gives

$$H^q(S(E)) \cong H^{q-2n+1}(\mathbb{CP}^n).$$

Therefore

$$H^q(S(E)) \cong \mathbb{Z}$$

when

$$q - 2n + 1$$

is even and lies between 0 and $2n$. Equivalently,

$$q = 2n+1, 2n+3, \ldots, 4n-1.$$

In all other degrees above $2n$, the cohomology vanishes.

Thus the cohomology groups of $S(E)$ are

$$H^q(S(E);\mathbb{Z}) \cong \begin{cases} \mathbb{Z}, & q = 0, 2, 4, \ldots, 2n-2, \\ 0, & q = 2n-1, \\ \mathbb{Z}/(n+1), & q = 2n, \\ \mathbb{Z}, & q = 2n+1, 2n+3, \ldots, 4n-1, \\ 0, & \text{otherwise.} \end{cases}$$

Step 7: Convert cohomology to homology

The total space $S(E)$ is a closed orientable manifold. Its dimension is

$$\dim S(E) = \dim \mathbb{CP}^n + \dim S^{2n-1} = 2n + (2n-1) = 4n-1.$$

By Poincaré duality,

$$H_k(S(E);\mathbb{Z}) \cong H^{4n-1-k}(S(E);\mathbb{Z}).$$

Using the cohomology computation above, we obtain

$$H_k(S(E);\mathbb{Z}) \cong \begin{cases} \mathbb{Z}, & k = 0, 2, 4, \dots, 2n-2, \\ \mathbb{Z}/(n+1), & k = 2n-1, \\ 0, & k = 2n, \\ \mathbb{Z}, & k = 2n+1, 2n+3, \dots, 4n-1, \\ 0, & \text{otherwise.} \end{cases}$$

Therefore

$$\boxed{H_k(S(T\mathbb{CP}^n);\mathbb{Z}) \cong \begin{cases} \mathbb{Z}, & k = 0, 2, 4, \dots, 2n-2, \\ \mathbb{Z}/(n+1), & k = 2n-1, \\ 0, & k = 2n, \\ \mathbb{Z}, & k = 2n+1, 2n+3, \dots, 4n-1, \\ 0, & \text{otherwise.} \end{cases}}$$

Check: The case $n = 1$

When

$$n = 1,$$

we have

$$\mathbb{CP}^1 \cong S^2.$$

The tangent bundle is TS^2, and its unit sphere bundle is

$$S(TS^2).$$

This is the unit tangent bundle of S^2, which is diffeomorphic to

$$SO(3) \cong \mathbb{RP}^3.$$

Our formula gives

$$H_0(S(TS^2)) \cong \mathbb{Z},$$
$$H_1(S(TS^2)) \cong \mathbb{Z}/2,$$
$$H_2(S(TS^2)) = 0,$$

and

$$H_3(S(TS^2)) \cong \mathbb{Z}.$$

These are exactly the homology groups of $\mathbb{RP}^3$. Hence the formula is consistent in this basic case.

12 Problems 6–8: Thom Classes, Bézout's Theorem, and Sphere Bundles

Throughout this section, cohomology is taken with integer coefficients unless otherwise stated.

13 Problem 6

Problem.

(a) Suppose u_i generates $H^{n_i}(\mathbb{R}^{n_i} \mid 0)$ for $i = 1, 2$. If

$$p_i : \mathbb{R}^{n_1} \times \mathbb{R}^{n_2} \longrightarrow \mathbb{R}^{n_i}$$

is the projection, show that

$$p_1^*(u_1) \smile p_2^*(u_2)$$

generates

$$H^{n_1+n_2}(\mathbb{R}^{n_1} \times \mathbb{R}^{n_2} \mid 0).$$

(b) Now suppose that

$$\pi_i : E_i \to B \qquad (i = 1, 2)$$

are vector bundles over B, and that U_i is a Thom class for E_i. If

$$p_i : E_1 \oplus E_2 \to E_i$$

is the projection, show that

$$p_1^*(U_1) \smile p_2^*(U_2)$$

is a Thom class for $E_1 \oplus E_2$.

(c) Deduce that $E_1 \oplus E_2$ can be oriented so that

$$e(E_1 \oplus E_2) = e(E_1) \smile e(E_2).$$

Solution.

Part (a)

We use the notation

$$H^k(\mathbb{R}^n \mid 0) = H^k(\mathbb{R}^n, \mathbb{R}^n \setminus \{0\}).$$

It is well known that

$$H^k(\mathbb{R}^n, \mathbb{R}^n \setminus \{0\}) \cong \begin{cases} \mathbb{Z}, & k = n, \\ 0, & k \neq n. \end{cases}$$

Hence

$$H^{n_i}(\mathbb{R}^{n_i} \mid 0) \cong \mathbb{Z}$$

for $i = 1, 2$, and

$$H^{n_1+n_2}(\mathbb{R}^{n_1} \times \mathbb{R}^{n_2} \mid 0) \cong \mathbb{Z}.$$

Let

$$u_i \in H^{n_i}(\mathbb{R}^{n_i}, \mathbb{R}^{n_i} \setminus \{0\})$$

be a generator. The product

$$p_1^*(u_1) \smile p_2^*(u_2)$$

is the local cohomology version of the external product

$$u_1 \times u_2.$$

Indeed, pulling back by the two projections and taking the cup product gives the class supported at the common zero

$$(0, 0) \in \mathbb{R}^{n_1} \times \mathbb{R}^{n_2}.$$

By the Künneth theorem for these local cohomology groups, the external product induces an isomorphism

$$H^{n_1}(\mathbb{R}^{n_1} \mid 0) \otimes H^{n_2}(\mathbb{R}^{n_2} \mid 0) \cong H^{n_1+n_2}(\mathbb{R}^{n_1} \times \mathbb{R}^{n_2} \mid 0).$$

Since both u_1 and u_2 are generators of infinite cyclic groups, their product is again a generator, up to sign.

Therefore

$$p_1^*(u_1) \smile p_2^*(u_2)$$

generates

$$H^{n_1+n_2}(\mathbb{R}^{n_1} \times \mathbb{R}^{n_2} \mid 0).$$

Thus

$$\boxed{p_1^*(u_1) \smile p_2^*(u_2) \text{ is a generator of } H^{n_1+n_2}(\mathbb{R}^{n_1} \times \mathbb{R}^{n_2} \mid 0).}$$

Part (b)

Let

$$\pi_i : E_i \to B$$

be vector bundles of ranks n_i. A Thom class for E_i is a class

$$U_i \in H^{n_i}(E_i, E_i \setminus B)$$

whose restriction to every fiber

$$(E_i)_b \cong \mathbb{R}^{n_i}$$

is a generator of

$$H^{n_i}((E_i)_b, (E_i)_b \setminus \{0\}) \cong \mathbb{Z}.$$

Now consider the direct sum bundle

$$E_1 \oplus E_2 \to B.$$

Its fiber over $b \in B$ is

$$(E_1 \oplus E_2)_b \cong (E_1)_b \oplus (E_2)_b \cong \mathbb{R}^{n_1} \oplus \mathbb{R}^{n_2}.$$

Let

$$p_i : E_1 \oplus E_2 \to E_i$$

be the bundle projection. Consider the class

$$U = p_1^*(U_1) \smile p_2^*(U_2).$$

This class has degree

$$n_1 + n_2.$$

Moreover, its support is along the zero section of $E_1 \oplus E_2$, so

$$U \in H^{n_1+n_2}(E_1 \oplus E_2, (E_1 \oplus E_2) \setminus B).$$

We must show that U restricts to a generator on each fiber.

Fix $b \in B$. Restricting U to the fiber gives

$$U|_{(E_1 \oplus E_2)_b} = p_1^*(U_1|_{(E_1)_b}) \smile p_2^*(U_2|_{(E_2)_b}).$$

Since U_i is a Thom class, each restriction

$$U_i|_{(E_i)_b}$$

is a generator of the local cohomology group of the fiber:

$$H^{n_i}((E_i)_b, (E_i)_b \setminus \{0\}).$$

By part (a), the cup product of these two pulled-back generators is a generator of

$$H^{n_1+n_2}\Big((E_1)_b \oplus (E_2)_b, ((E_1)_b \oplus (E_2)_b) \setminus \{0\}\Big).$$

Therefore the restriction of

$$p_1^*(U_1) \smile p_2^*(U_2)$$

to every fiber is a generator. Hence this class is a Thom class for $E_1 \oplus E_2$.

Thus

$$\boxed{p_1^*(U_1) \smile p_2^*(U_2) \text{ is a Thom class for } E_1 \oplus E_2.}$$

Part (c)

The choice of a Thom class determines an orientation of a vector bundle. By part (b), the class

$$U = p_1^*(U_1) \smile p_2^*(U_2)$$

is a Thom class for $E_1 \oplus E_2$. We orient $E_1 \oplus E_2$ using this Thom class.

Let

$$s : B \to E_1 \oplus E_2$$

be the zero section of $E_1 \oplus E_2$. By definition, the Euler class of an oriented vector bundle is the pullback of its Thom class along the zero section. Therefore

$$e(E_1 \oplus E_2) = s^*(U).$$

Using the formula for U, we get

$$e(E_1 \oplus E_2) = s^*\Big(p_1^*(U_1) \smile p_2^*(U_2)\Big).$$

By naturality of cup products,

$$e(E_1 \oplus E_2) = s^*p_1^*(U_1) \smile s^*p_2^*(U_2).$$

But

$$p_i \circ s : B \to E_i$$

is precisely the zero section of E_i. Hence

$$s^*p_i^*(U_i) = (p_i \circ s)^*(U_i) = e(E_i).$$

Therefore

$$e(E_1 \oplus E_2) = e(E_1) \smile e(E_2).$$

Thus

$$\boxed{e(E_1 \oplus E_2) = e(E_1) \smile e(E_2).}$$

14 Problem 7

Problem. If

$$p \in \mathbb{C}[z_0, z_1, z_2]$$

is a homogeneous polynomial, define

$$V_p = \{[z_0 : z_1 : z_2] \in \mathbb{CP}^2 \mid p(z_0, z_1, z_2) = 0\}.$$

If p and q are chosen so that V_p and V_q are embedded submanifolds of $\mathbb{CP}^2$ which intersect transversely, show that V_p and V_q intersect in precisely

$$(\deg p)(\deg q)$$

points.

Solution.

Let

$$d = \deg p, \qquad e = \deg q.$$

The zero set V_p is a complex hypersurface in $\mathbb{CP}^2$, hence a complex curve. Since V_p is assumed to be an embedded submanifold, it defines a homology class

$$[V_p] \in H_2(\mathbb{CP}^2; \mathbb{Z}).$$

Let

$$a \in H^2(\mathbb{CP}^2; \mathbb{Z})$$

be the standard generator. Then

$$H^*(\mathbb{CP}^2; \mathbb{Z}) \cong \mathbb{Z}[a]/(a^3), \qquad \deg a = 2,$$

and

$$\langle a^2, [\mathbb{CP}^2] \rangle = 1.$$

A homogeneous polynomial of degree d defines a section of the complex line bundle

$$\mathcal{O}(d) \to \mathbb{CP}^2.$$

Its zero set V_p has Poincaré dual equal to the first Chern class of $\mathcal{O}(d)$:

$$[V_p]^\vee = c_1(\mathcal{O}(d)).$$

Since

$$c_1(\mathcal{O}(d)) = d\, a,$$

we obtain

$$[V_p]^\vee = d\, a.$$

Similarly,

$$[V_q]^\vee = e\, a.$$

The algebraic intersection number of V_p and V_q is therefore

$$V_p \cdot V_q = \langle [V_p]^\vee \smile [V_q]^\vee, [\mathbb{CP}^2] \rangle.$$

Substituting the Poincaré dual classes gives

$$V_p \cdot V_q = \langle (d\, a) \smile (e\, a), [\mathbb{CP}^2] \rangle.$$

Hence

$$V_p \cdot V_q = de \langle a^2, [\mathbb{CP}^2] \rangle.$$

Since

$$\langle a^2, [\mathbb{CP}^2] \rangle = 1,$$

we get

$$V_p \cdot V_q = de.$$

Because V_p and V_q are complex submanifolds of the complex surface $\mathbb{CP}^2$, every transverse intersection point has local intersection number $+1$. Therefore the algebraic intersection number is equal to the actual number of intersection points.

Thus

$$|V_p \cap V_q| = de = (\deg p)(\deg q).$$

Therefore

$$\boxed{|V_p \cap V_q| = (\deg p)(\deg q).}$$

This is the smooth transverse case of Bézout's theorem for projective plane curves.

15 Problem 8

Problem. Let E be the tangent bundle of $\mathbb{CP}^n$. Compute

$$H_*(S(E)).$$

Solution.

Let

$$B = \mathbb{CP}^n, \qquad E = T\mathbb{CP}^n.$$

Since $\mathbb{CP}^n$ is a complex manifold of complex dimension n, its tangent bundle $T\mathbb{CP}^n$ is a complex vector bundle of complex rank n. As a real vector bundle, it has rank

$$2n.$$

Therefore the sphere bundle

$$S(E) = S(T\mathbb{CP}^n)$$

has fiber

$$S^{2n-1}.$$

So we have an oriented sphere bundle

$$S^{2n-1} \longrightarrow S(T\mathbb{CP}^n) \xrightarrow{\pi} \mathbb{CP}^n.$$

Step 1: Cohomology of $\mathbb{CP}^n$

The cohomology ring of complex projective space is

$$H^*(\mathbb{CP}^n;\mathbb{Z}) \cong \mathbb{Z}[x]/(x^{n+1}), \qquad \deg x = 2.$$

Hence

$$H^{2k}(\mathbb{CP}^n;\mathbb{Z}) \cong \mathbb{Z} \qquad (0 \le k \le n),$$

and

$$H^{2k+1}(\mathbb{CP}^n;\mathbb{Z}) = 0.$$

Step 2: Euler class of $T\mathbb{CP}^n$

For $\mathbb{CP}^n$, the total Chern class of the tangent bundle is

$$c(T\mathbb{CP}^n) = (1+x)^{n+1}.$$

Therefore the top Chern class is

$$c_n(T\mathbb{CP}^n) = (n+1)x^n.$$

Since $T\mathbb{CP}^n$ is a complex vector bundle, the Euler class of its underlying oriented real bundle is the top Chern class:

$$e(T\mathbb{CP}^n) = c_n(T\mathbb{CP}^n).$$

Hence

$$e(E) = e(T\mathbb{CP}^n) = (n+1)x^n.$$

Step 3: The Gysin sequence

For an oriented sphere bundle coming from an oriented real vector bundle of rank $2n$, the Gysin sequence has the form

$$\cdots \longrightarrow H^{q-2n}(\mathbb{CP}^n) \xrightarrow{\smile e(E)} H^q(\mathbb{CP}^n) \longrightarrow H^q(S(E)) \longrightarrow H^{q-2n+1}(\mathbb{CP}^n) \xrightarrow{\smile e(E)} H^{q+1}(\mathbb{CP}^n) \longrightarrow \cdots.$$

Since

$$H^{\mathrm{odd}}(\mathbb{CP}^n) = 0,$$

this sequence simplifies significantly.

Step 4: Degrees below $2n-1$

For

$$q < 2n-1,$$

we have

$$H^{q-2n}(\mathbb{CP}^n) = 0$$

and

$$H^{q-2n+1}(\mathbb{CP}^n) = 0.$$

Therefore the Gysin sequence gives

$$H^q(S(E)) \cong H^q(\mathbb{CP}^n).$$

Hence

$$H^q(S(E)) \cong \mathbb{Z}$$

for even q satisfying

$$0 \leq q \leq 2n-2,$$

and

$$H^q(S(E)) = 0$$

for odd $q < 2n-1$.

Step 5: The middle part

Now take

$$q = 2n-1$$

and

$$q = 2n.$$

The relevant portion of the Gysin sequence is

$$0 \longrightarrow H^{2n-1}(S(E)) \longrightarrow H^0(\mathbb{CP}^n) \xrightarrow{\smile e(E)} H^{2n}(\mathbb{CP}^n) \longrightarrow H^{2n}(S(E)) \longrightarrow 0.$$

Since

$$H^0(\mathbb{CP}^n) \cong \mathbb{Z}, \qquad H^{2n}(\mathbb{CP}^n) \cong \mathbb{Z},$$

and
$$e(E) = (n+1)x^n,$$
the map
$$H^0(\mathbb{CP}^n) \longrightarrow H^{2n}(\mathbb{CP}^n)$$
is multiplication by $n+1$. Therefore the exact sequence becomes
$$0 \longrightarrow H^{2n-1}(S(E)) \longrightarrow \mathbb{Z} \xrightarrow{\times(n+1)} \mathbb{Z} \longrightarrow H^{2n}(S(E)) \longrightarrow 0.$$

The map
$$\mathbb{Z} \xrightarrow{\times(n+1)} \mathbb{Z}$$
is injective, so
$$H^{2n-1}(S(E)) = 0.$$
Its cokernel is
$$\mathbb{Z}/(n+1).$$
Hence
$$H^{2n}(S(E)) \cong \mathbb{Z}/(n+1).$$

Step 6: Degrees above $2n$

For
$$q > 2n,$$
we have
$$H^q(\mathbb{CP}^n) = 0.$$
The Gysin sequence then gives
$$H^q(S(E)) \cong H^{q-2n+1}(\mathbb{CP}^n).$$

Therefore
$$H^q(S(E)) \cong \mathbb{Z}$$
when
$$q - 2n + 1$$
is even and lies between 0 and $2n$. Equivalently,
$$q = 2n+1, 2n+3, \ldots, 4n-1.$$

In all other degrees above $2n$, the cohomology vanishes.

Thus the cohomology groups of $S(E)$ are
$$H^q(S(E);\mathbb{Z}) \cong \begin{cases} \mathbb{Z}, & q = 0, 2, 4, \ldots, 2n-2, \\ 0, & q = 2n-1, \\ \mathbb{Z}/(n+1), & q = 2n, \\ \mathbb{Z}, & q = 2n+1, 2n+3, \ldots, 4n-1, \\ 0, & \text{otherwise.} \end{cases}$$

Step 7: Convert cohomology to homology

The total space $S(E)$ is a closed orientable manifold. Its dimension is

$$\dim S(E) = \dim \mathbb{CP}^n + \dim S^{2n-1} = 2n + (2n-1) = 4n - 1.$$

By Poincaré duality,

$$H_k(S(E);\mathbb{Z}) \cong H^{4n-1-k}(S(E);\mathbb{Z}).$$

Using the cohomology computation above, we obtain

$$H_k(S(E);\mathbb{Z}) \cong \begin{cases} \mathbb{Z}, & k = 0, 2, 4, \dots, 2n-2, \\ \mathbb{Z}/(n+1), & k = 2n-1, \\ 0, & k = 2n, \\ \mathbb{Z}, & k = 2n+1, 2n+3, \dots, 4n-1, \\ 0, & \text{otherwise.} \end{cases}$$

Therefore

$$\boxed{H_k(S(T\mathbb{CP}^n);\mathbb{Z}) \cong \begin{cases} \mathbb{Z}, & k = 0, 2, 4, \dots, 2n-2, \\ \mathbb{Z}/(n+1), & k = 2n-1, \\ 0, & k = 2n, \\ \mathbb{Z}, & k = 2n+1, 2n+3, \dots, 4n-1, \\ 0, & \text{otherwise.} \end{cases}}$$

Check: The case $n = 1$

When

$$n = 1,$$

we have

$$\mathbb{CP}^1 \cong S^2.$$

The tangent bundle is TS^2, and its unit sphere bundle is

$$S(TS^2).$$

This is the unit tangent bundle of S^2, which is diffeomorphic to

$$SO(3) \cong \mathbb{RP}^3.$$

Our formula gives

$$H_0(S(TS^2)) \cong \mathbb{Z},$$
$$H_1(S(TS^2)) \cong \mathbb{Z}/2,$$
$$H_2(S(TS^2)) = 0,$$

and

$$H_3(S(TS^2)) \cong \mathbb{Z}.$$

These are exactly the homology groups of $\mathbb{RP}^3$. Hence the formula is consistent in this basic case.

16 Theory for Problems 6–8: Thom Classes, Euler Classes, Bézout's Theorem, and Gysin Sequences

These three problems are connected by one main idea:

$$\text{geometric data} \quad \longrightarrow \quad \text{cohomology classes} \quad \longrightarrow \quad \text{intersection or characteristic-class computations.}$$

Problem 6 is about Thom classes and Euler classes. Problem 7 is about projective hypersurfaces and intersection numbers. Problem 8 is about sphere bundles and the Gysin sequence.

16.1 Local Cohomology at a Point

In Problem 6(a), the notation

$$H^n(\mathbb{R}^n \mid 0)$$

means relative cohomology:

$$H^n(\mathbb{R}^n, \mathbb{R}^n \setminus \{0\}).$$

This group measures cohomology localized at the origin. Since

$$\mathbb{R}^n/(\mathbb{R}^n \setminus \{0\})$$

has the homotopy type of S^n, we obtain

$$H^k(\mathbb{R}^n, \mathbb{R}^n \setminus \{0\}) \cong \widetilde{H}^k(S^n).$$

Therefore

$$H^k(\mathbb{R}^n \mid 0) \cong \begin{cases} \mathbb{Z}, & k = n, \\ 0, & k \neq n. \end{cases}$$

Thus the group

$$H^n(\mathbb{R}^n \mid 0)$$

has exactly two generators, differing by sign. Choosing one generator is the same as choosing an orientation of $\mathbb{R}^n$.

16.2 Product of Local Orientation Classes

Suppose

$$u_1 \in H^{n_1}(\mathbb{R}^{n_1} \mid 0), \qquad u_2 \in H^{n_2}(\mathbb{R}^{n_2} \mid 0)$$

are generators.

The product space is

$$\mathbb{R}^{n_1} \times \mathbb{R}^{n_2} \cong \mathbb{R}^{n_1+n_2}.$$

The origin in the product is

$$(0, 0).$$

Let

$$p_i : \mathbb{R}^{n_1} \times \mathbb{R}^{n_2} \to \mathbb{R}^{n_i}$$

be the projections. Pulling back the local classes and multiplying gives

$$p_1^*(u_1) \smile p_2^*(u_2).$$

This class lies in

$$H^{n_1+n_2}(\mathbb{R}^{n_1} \times \mathbb{R}^{n_2} \mid 0).$$

The key fact is:

$$\text{orientation class of a product} = \text{product of orientation classes.}$$

Therefore

$$p_1^*(u_1) \smile p_2^*(u_2)$$

is a generator of

$$H^{n_1+n_2}(\mathbb{R}^{n_1} \times \mathbb{R}^{n_2} \mid 0).$$

Geometrically, this says that if $\mathbb{R}^{n_1}$ and $\mathbb{R}^{n_2}$ are oriented, then their product orientation gives an orientation of

$$\mathbb{R}^{n_1+n_2}.$$

16.3 Thom Classes

Let

$$\pi : E \to B$$

be an oriented real vector bundle of rank r. Let

$$s : B \to E$$

be the zero section. We identify B with its image $s(B) \subset E$.

A *Thom class* of E is a cohomology class

$$U \in H^r(E, E \setminus B)$$

such that for every point $b \in B$, the restriction of U to the fiber

$$E_b \cong \mathbb{R}^r$$

is a generator of

$$H^r(E_b, E_b \setminus \{0\}) \cong \mathbb{Z}.$$

Thus a Thom class is a family of local orientation classes, one in each fiber, varying continuously over the base.

The Thom class is the cohomological expression of the orientation of the vector bundle.

16.4 The Thom Isomorphism

If $E \to B$ is an oriented rank-r vector bundle with Thom class U, then cup product with U gives the *Thom isomorphism*:

$$H^k(B) \cong H^{k+r}(E, E \setminus B).$$

More precisely, the isomorphism is

$$\alpha \longmapsto \pi^*(\alpha) \smile U.$$

So the Thom class shifts cohomology by the rank of the bundle. This is why Thom classes are important: they convert information on the base into relative cohomology information on the total space of the bundle.

16.5 Thom Class of a Direct Sum Bundle

Suppose

$$E_1 \to B, \qquad E_2 \to B$$

are oriented vector bundles of ranks n_1 and n_2, with Thom classes

$$U_1 \in H^{n_1}(E_1, E_1 \setminus B),$$

and

$$U_2 \in H^{n_2}(E_2, E_2 \setminus B).$$

The direct sum bundle is

$$E_1 \oplus E_2 \to B.$$

Its fiber over $b \in B$ is

$$(E_1 \oplus E_2)_b = (E_1)_b \oplus (E_2)_b.$$

Therefore the local orientation class of the fiber should be the product of the local orientation classes of the two factors. This is precisely why

$$p_1^*(U_1) \smile p_2^*(U_2)$$

is a Thom class for

$$E_1 \oplus E_2.$$

This is the bundle version of the local calculation from Problem 6(a). The direct sum orientation is therefore the product orientation:

$$\mathrm{or}(E_1 \oplus E_2) = \mathrm{or}(E_1)\,\mathrm{or}(E_2).$$

16.6 Euler Classes

Let

$$\pi : E \to B$$

be an oriented rank-r vector bundle with Thom class

$$U \in H^r(E, E \setminus B).$$

The Euler class of E is defined by pulling the Thom class back along the zero section:

$$e(E) = s^*(U) \in H^r(B).$$

Thus the Euler class is the restriction of the Thom class back to the base.

Geometrically, the Euler class measures the obstruction to finding a nowhere-zero section. Intuitively,

$$e(E) = 0$$

means that the bundle may admit a nowhere-zero section, while

$$e(E) \neq 0$$

detects twisting that prevents such a section.

For example, the tangent bundle of S^2 has a nonzero Euler class:

$$e(TS^2) \neq 0.$$

This corresponds to the hairy ball theorem.

16.7 Multiplicativity of the Euler Class

For oriented vector bundles

$$E_1 \to B, \qquad E_2 \to B,$$

Problem 6 proves the formula

$$e(E_1 \oplus E_2) = e(E_1) \smile e(E_2).$$

The reason is as follows. The Thom class of $E_1 \oplus E_2$ is

$$U_{E_1 \oplus E_2} = p_1^*(U_1) \smile p_2^*(U_2).$$

The Euler class is the pullback of the Thom class along the zero section. Hence

$$e(E_1 \oplus E_2) = s^*\Big(p_1^*(U_1) \smile p_2^*(U_2)\Big).$$

By naturality of cup products,

$$e(E_1 \oplus E_2) = s^* p_1^*(U_1) \smile s^* p_2^*(U_2).$$

But

$$s^* p_i^*(U_i) = e(E_i).$$

Therefore

$$e(E_1 \oplus E_2) = e(E_1) \smile e(E_2).$$

This is one of the fundamental formal properties of the Euler class.

16.8 Projective Space and the Hyperplane Class

Problem 7 takes place in complex projective space

$$\mathbb{CP}^2.$$

Its cohomology ring is

$$H^*(\mathbb{CP}^2;\mathbb{Z}) \cong \mathbb{Z}[a]/(a^3), \qquad \deg a = 2.$$

The class

$$a \in H^2(\mathbb{CP}^2;\mathbb{Z})$$

is called the *hyperplane class.* Geometrically, a is Poincaré dual to a projective line

$$\mathbb{CP}^1 \subset \mathbb{CP}^2.$$

Since two projective lines intersect in one point, we have

$$\langle a^2, [\mathbb{CP}^2]\rangle = 1.$$

This relation is the cohomological foundation of Bézout's theorem in $\mathbb{CP}^2$.

16.9 Homogeneous Polynomials and Projective Curves

A homogeneous polynomial

$$p(z_0, z_1, z_2)$$

of degree d defines a projective curve

$$V_p = \{[z_0 : z_1 : z_2] \in \mathbb{CP}^2 \mid p(z_0, z_1, z_2) = 0\}.$$

Homogeneity is necessary because projective coordinates are defined only up to nonzero scalar multiplication:

$$[z_0 : z_1 : z_2] = [\lambda z_0 : \lambda z_1 : \lambda z_2].$$

If p is homogeneous of degree d, then

$$p(\lambda z_0, \lambda z_1, \lambda z_2) = \lambda^d p(z_0, z_1, z_2).$$

Therefore the condition

$$p(z_0, z_1, z_2) = 0$$

is well-defined on projective space.

16.10 Degree and Poincaré Dual Class

A degree-d homogeneous polynomial defines a section of the line bundle

$$\mathcal{O}(d) \to \mathbb{CP}^2.$$

The zero set of a transverse section has Poincaré dual equal to the Euler class of the bundle. Since $\mathcal{O}(d)$ is a complex line bundle, its Euler class equals its first Chern class:

$$e(\mathcal{O}(d)) = c_1(\mathcal{O}(d)).$$

We have

$$c_1(\mathcal{O}(d)) = d\,a.$$

Therefore

$$[V_p]^\vee = d\,a.$$

Similarly, if

$$\deg q = e,$$

then

$$[V_q]^\vee = e\,a.$$

Thus the degree of the polynomial becomes the coefficient of the hyperplane class.

16.11 Intersection Number in $\mathbb{CP}^2$

The intersection number of two oriented submanifolds is computed by multiplying their Poincaré dual classes. Therefore

$$V_p \cdot V_q = \langle [V_p]^\vee \smile [V_q]^\vee, [\mathbb{CP}^2] \rangle.$$

Using

$$[V_p]^\vee = d\,a, \qquad [V_q]^\vee = e\,a,$$

we get

$$V_p \cdot V_q = \langle (d\,a) \smile (e\,a), [\mathbb{CP}^2] \rangle.$$

Therefore

$$V_p \cdot V_q = de\langle a^2, [\mathbb{CP}^2] \rangle.$$

Since

$$\langle a^2, [\mathbb{CP}^2] \rangle = 1,$$

we obtain

$$V_p \cdot V_q = de.$$

Thus the algebraic intersection number is

$$(\deg p)(\deg q).$$

16.12 Why Transverse Complex Intersections Count Positively

For two real oriented submanifolds, intersection points can have signs

$$+1 \qquad \text{or} \qquad -1.$$

Therefore algebraic intersection number may differ from the actual number of intersection points.

But in complex geometry, transverse intersections of complex submanifolds always have positive local intersection number.

The reason is that complex vector spaces have canonical orientations. If two complex submanifolds intersect transversely, their tangent spaces fit together compatibly with the complex orientation of the ambient space.

Therefore every transverse intersection point contributes

$$+1.$$

Hence in Problem 7,

$$\text{algebraic intersection number} = \text{actual number of intersection points.}$$

So

$$|V_p \cap V_q| = (\deg p)(\deg q).$$

This is the clean transverse version of Bézout's theorem.

16.13 Bézout's Theorem: Conceptual Form

Problem 7 is a special case of Bézout's theorem.

For two projective plane curves

$$V_p, V_q \subset \mathbb{CP}^2$$

of degrees d and e, if they have no common component, then their intersection number counted with multiplicity is

$$de.$$

If the curves are smooth and intersect transversely, then every multiplicity is 1, so the actual number of intersection points is

$$de.$$

For example:

$$\begin{aligned} \text{line} \cap \text{line}: &\qquad 1 \cdot 1 = 1, \\ \text{line} \cap \text{conic}: &\qquad 1 \cdot 2 = 2, \\ \text{conic} \cap \text{conic}: &\qquad 2 \cdot 2 = 4, \end{aligned}$$

and

$$\text{cubic} \cap \text{conic}: \qquad 3 \cdot 2 = 6.$$

16.14 Sphere Bundles

In Problem 8, E is the tangent bundle of $\mathbb{CP}^n$:

$$E = T\mathbb{CP}^n.$$

The sphere bundle

$$S(E)$$

is the bundle whose fiber over a point $b \in \mathbb{CP}^n$ is the unit sphere in the vector space E_b.

Since $\mathbb{CP}^n$ has complex dimension n, its tangent bundle has complex rank n, hence real rank

$$2n.$$

Therefore the fiber of

$$S(T\mathbb{CP}^n) \to \mathbb{CP}^n$$

is

$$S^{2n-1}.$$

Thus we have a sphere bundle

$$S^{2n-1} \longrightarrow S(T\mathbb{CP}^n) \longrightarrow \mathbb{CP}^n.$$

The total space has dimension

$$\dim S(T\mathbb{CP}^n) = \dim \mathbb{CP}^n + \dim S^{2n-1} = 2n + (2n-1) = 4n - 1.$$

16.15 Chern Classes of $T\mathbb{CP}^n$

The cohomology ring of $\mathbb{CP}^n$ is

$$H^*(\mathbb{CP}^n; \mathbb{Z}) \cong \mathbb{Z}[x]/(x^{n+1}), \qquad \deg x = 2.$$

The tangent bundle of complex projective space has total Chern class

$$c(T\mathbb{CP}^n) = (1+x)^{n+1}.$$

The top Chern class is therefore

$$c_n(T\mathbb{CP}^n) = (n+1)x^n.$$

For a complex vector bundle, the Euler class of the underlying real bundle equals the top Chern class:

$$e(E_{\mathbb{R}}) = c_n(E).$$

Thus

$$e(T\mathbb{CP}^n) = (n+1)x^n.$$

This Euler class is the crucial input in Problem 8.

16.16 The Gysin Sequence

Let
$$S^{r-1} \to S(E) \xrightarrow{\pi} B$$
be the sphere bundle of an oriented real rank-r vector bundle
$$E \to B.$$

Then there is a long exact sequence called the *Gysin sequence*:
$$\cdots \to H^{q-r}(B) \xrightarrow{\smile e(E)} H^q(B) \to H^q(S(E)) \to H^{q-r+1}(B) \xrightarrow{\smile e(E)} H^{q+1}(B) \to \cdots .$$

This sequence relates
$$H^*(B), \qquad H^*(S(E)), \qquad e(E).$$

Therefore the topology of the sphere bundle is controlled by the Euler class of the original vector bundle.

In Problem 8,
$$r = 2n,$$
so the Gysin sequence becomes
$$\cdots \to H^{q-2n}(\mathbb{CP}^n) \xrightarrow{\smile e(E)} H^q(\mathbb{CP}^n) \to H^q(S(E)) \to H^{q-2n+1}(\mathbb{CP}^n) \to \cdots .$$

16.17 Why the Gysin Sequence Simplifies for $\mathbb{CP}^n$

The cohomology of $\mathbb{CP}^n$ is concentrated in even degrees:
$$H^{2k}(\mathbb{CP}^n) \cong \mathbb{Z}, \qquad 0 \le k \le n,$$
and
$$H^{2k+1}(\mathbb{CP}^n) = 0.$$

Because all odd cohomology groups vanish, many terms in the Gysin sequence are zero. This makes the computation of
$$H^*(S(T\mathbb{CP}^n))$$
almost automatic.

The only nontrivial map is the Euler-class multiplication
$$H^0(\mathbb{CP}^n) \xrightarrow{\smile e(E)} H^{2n}(\mathbb{CP}^n).$$

Since
$$e(E) = (n+1)x^n,$$
this map is
$$\mathbb{Z} \xrightarrow{\times(n+1)} \mathbb{Z}.$$

Its cokernel is
$$\mathbb{Z}/(n+1).$$

That is exactly where the torsion group in the answer comes from.

16.18 The Torsion Group $\mathbb{Z}/(n+1)$

The appearance of
$$\mathbb{Z}/(n+1)$$
in
$$H_{2n-1}(S(T\mathbb{CP}^n))$$
comes directly from the Euler class
$$e(T\mathbb{CP}^n) = (n+1)x^n.$$

In the Gysin sequence, one obtains the map
$$\mathbb{Z} \xrightarrow{\times(n+1)} \mathbb{Z}.$$

The quotient is
$$\operatorname{coker}(\times(n+1)) = \mathbb{Z}/(n+1).$$

Thus the torsion in the homology of the sphere bundle records the Euler characteristic of the base:
$$\chi(\mathbb{CP}^n) = n+1.$$

Indeed, for a closed oriented $2n$-manifold M,
$$\langle e(TM), [M] \rangle = \chi(M).$$

For $M = \mathbb{CP}^n$, this gives
$$\chi(\mathbb{CP}^n) = n+1.$$

16.19 Unit Tangent Bundle Viewpoint

Since
$$S(T\mathbb{CP}^n)$$
is the sphere bundle of the tangent bundle, it is the *unit tangent bundle* of $\mathbb{CP}^n$.

A point of
$$S(T\mathbb{CP}^n)$$
is a pair
$$(x, v),$$
where
$$x \in \mathbb{CP}^n$$
and
$$v \in T_x\mathbb{CP}^n$$
is a unit tangent vector.

Thus
$$S(T\mathbb{CP}^n) = \{(x, v) \mid x \in \mathbb{CP}^n,\ v \in T_x\mathbb{CP}^n,\ \|v\| = 1\}.$$

For $n = 1$,

$$\mathbb{CP}^1 \cong S^2.$$

Then

$$S(T\mathbb{CP}^1) = S(TS^2)$$

is the unit tangent bundle of the sphere. This space is

$$SO(3) \cong \mathbb{RP}^3.$$

So the formula from Problem 8 gives

$$H_*(S(TS^2)) = H_*(\mathbb{RP}^3),$$

which is a useful consistency check.

16.20 Summary of the Three Problems

The three problems form a chain of ideas:

$$\text{Thom class} \quad \longrightarrow \quad \text{Euler class} \quad \longrightarrow \quad \text{intersection theory and Gysin sequences.}$$

Problem 6 proves the multiplicative behavior:

$$U_{E_1 \oplus E_2} = p_1^*(U_1) \smile p_2^*(U_2),$$

and therefore

$$e(E_1 \oplus E_2) = e(E_1) \smile e(E_2).$$

Problem 7 uses the same philosophy:

$$\text{zero set of a section} \quad \longleftrightarrow \quad \text{Euler/Chern class.}$$

A degree-d projective curve has Poincaré dual

$$d\,a.$$

Therefore two curves of degrees d and e intersect in

$$de$$

points when the intersection is transverse.

Problem 8 uses the Euler class inside the Gysin sequence. Since

$$e(T\mathbb{CP}^n) = (n+1)x^n,$$

the key map is

$$\mathbb{Z} \xrightarrow{\times(n+1)} \mathbb{Z}.$$

That map creates the torsion group

$$\mathbb{Z}/(n+1).$$

Hence:

$$\boxed{\text{Problem 6: Thom classes multiply.}}$$

$$\boxed{\text{Problem 7: degree becomes intersection number.}}$$

$$\boxed{\text{Problem 8: Euler class controls the sphere bundle via Gysin.}}$$

16.21 Figures and Tables for Problems 6–8

Remark

The following figures and tables summarize the main ideas of Problems 6–8 in a compact form suitable for a narrow page layout.

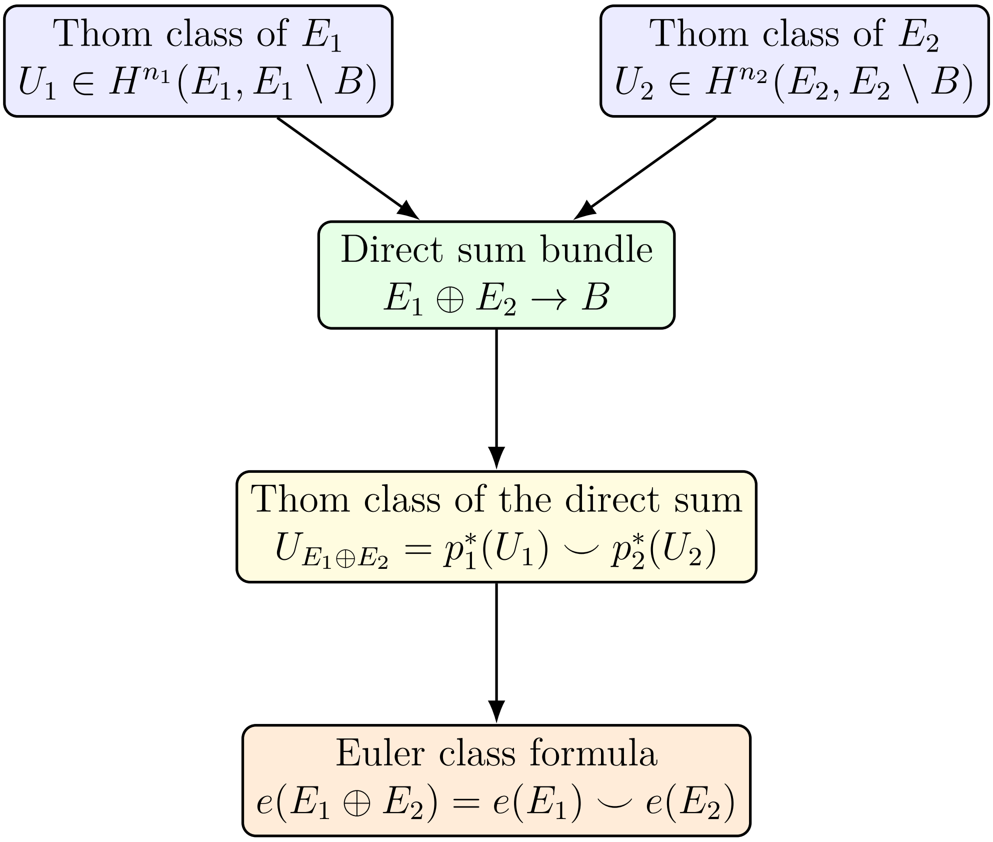


Figure 7: From Thom classes to the Euler class formula for a direct sum bundle.
Source: own material.

Table 1: Summary of the main identities from Problem 6.*Source: own material.*

Object	**Cohomology class**	**Main formula**
E_1	$U_1 \in H^{n_1}(E_1, E_1 \setminus B)$	Thom class of E_1
E_2	$U_2 \in H^{n_2}(E_2, E_2 \setminus B)$	Thom class of E_2
$E_1 \oplus E_2$	$p_1^*(U_1) \smile p_2^*(U_2)$	$U_{E_1 \oplus E_2} = p_1^*(U_1) \smile p_2^*(U_2)$
$E_1 \oplus E_2$	$e(E_1 \oplus E_2)$	$e(E_1 \oplus E_2) = e(E_1) \smile e(E_2)$

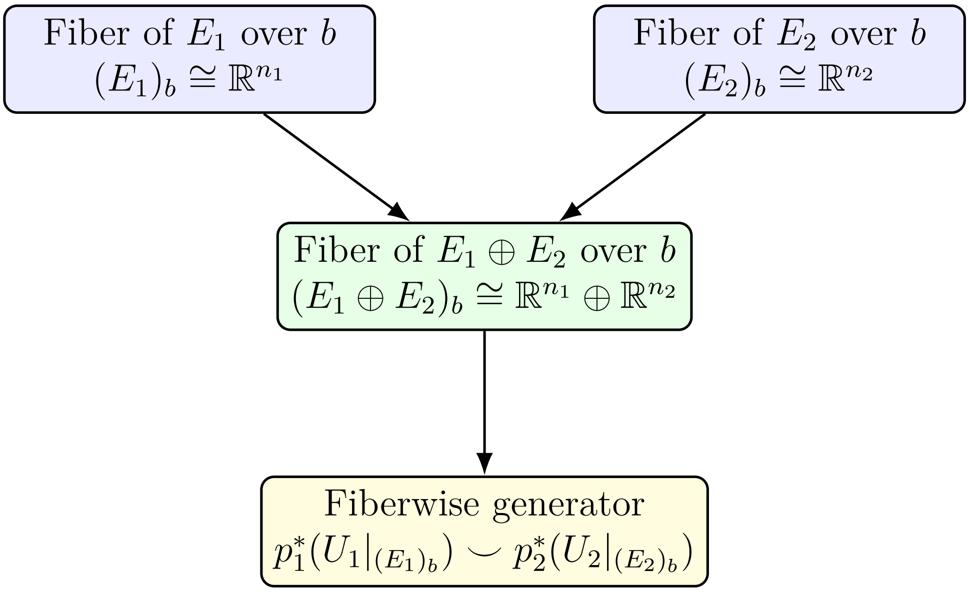


Figure 8: Fiberwise picture behind the Thom class of $E_1 \oplus E_2$.*Source: own material.*

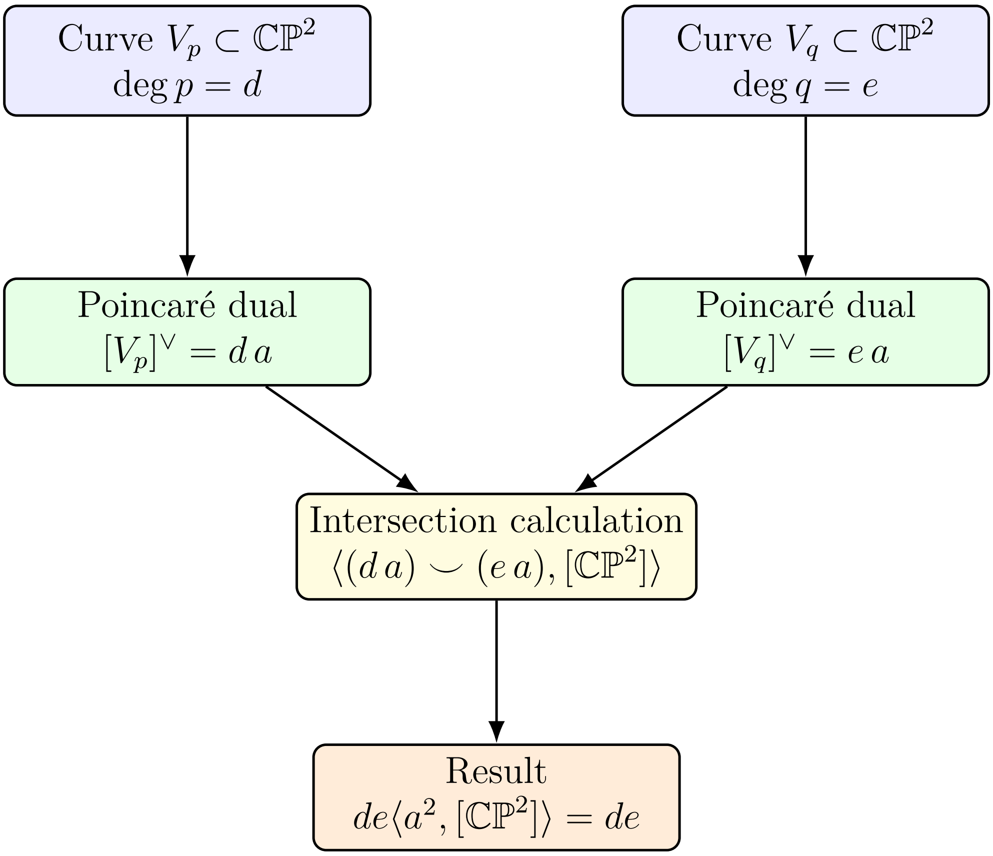


Figure 9: Cohomological mechanism behind Bézout's theorem in $\mathbb{CP}^2$.
Source: own material.

Table 2: Examples illustrating Bézout's theorem in $\mathbb{CP}^2$. *Source: own material.*

Curve V_p	$\deg p$	**Curve** V_q	$\lvert V_p \cap V_q\rvert$
line	1	line	1
line	1	conic	2
conic	2	conic	4
cubic	3	line	3
cubic	3	conic	6

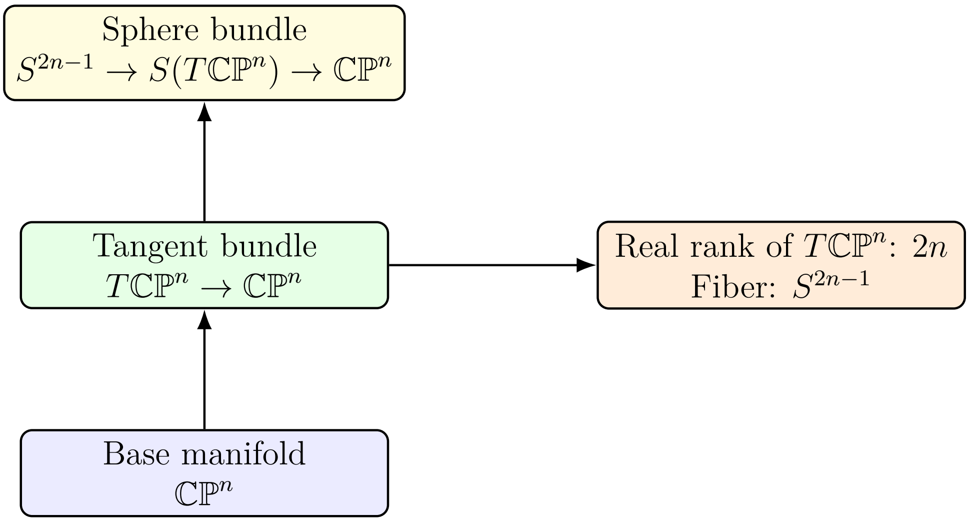


Figure 10: Geometric setup for the sphere bundle $S(T\mathbb{CP}^n)$. *Source: own material.*

Table 3: Cohomology of $S(T\mathbb{CP}^n)$. *Source: own material.*

Degree q	$\mathbf{H^q(S(T\mathbb{CP}^n);\mathbb{Z})}$
$0, 2, 4, \ldots, 2n-2$	$\mathbb{Z}$
$2n-1$	0
$2n$	$\mathbb{Z}/(n+1)$
$2n+1, 2n+3, \ldots, 4n-1$	$\mathbb{Z}$
otherwise	0

Table 4: Homology of $S(T\mathbb{CP}^n)$. *Source: own material.*

Degree k	$\mathbf{H_k(S(T\mathbb{CP}^n);\mathbb{Z})}$
$0, 2, 4, \ldots, 2n-2$	$\mathbb{Z}$
$2n-1$	$\mathbb{Z}/(n+1)$
$2n$	0
$2n+1, 2n+3, \ldots, 4n-1$	$\mathbb{Z}$
otherwise	0

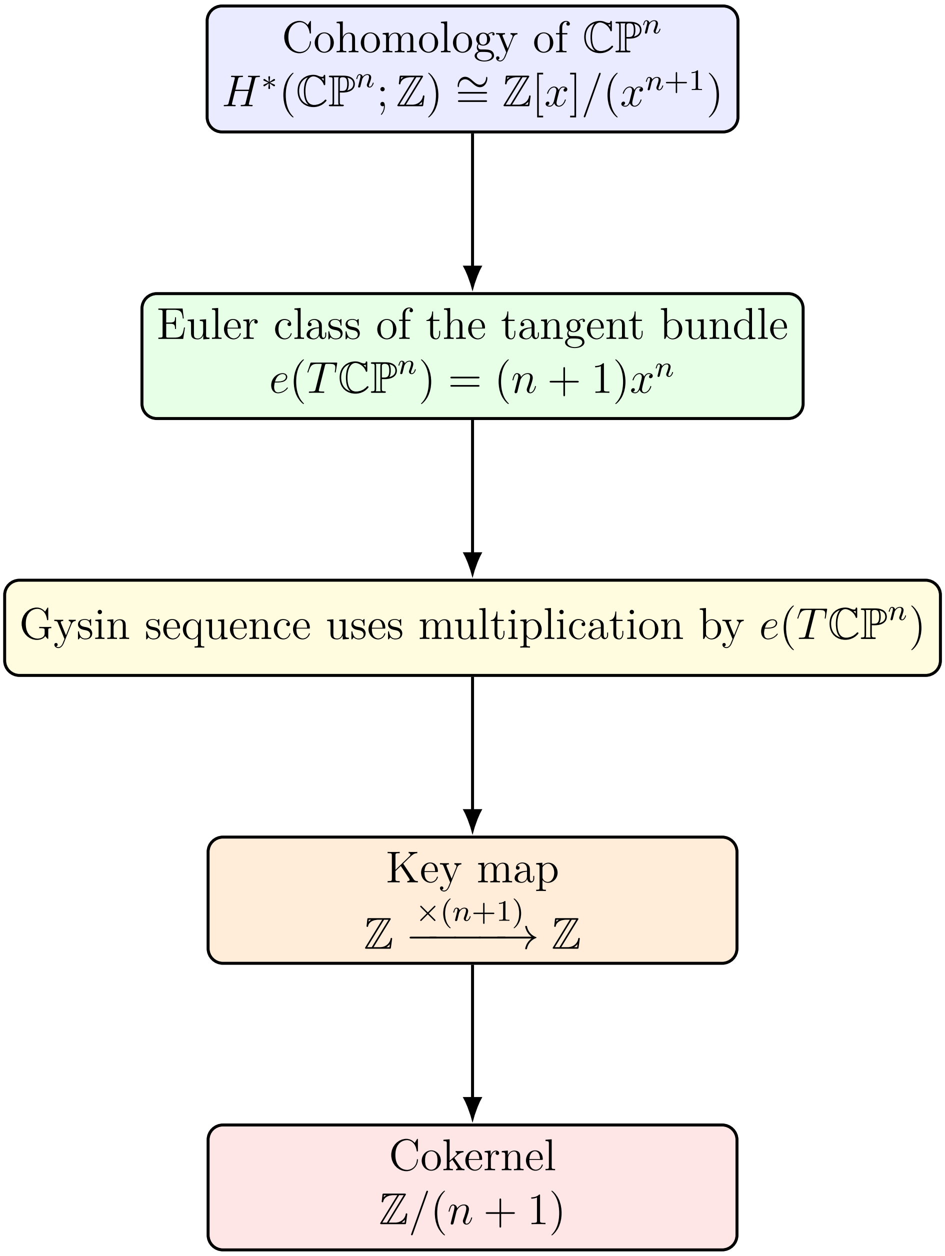


Figure 11: How the Gysin sequence produces the torsion term in $H_*(S(T\mathbb{CP}^n))$.
Source: own material.

Table 5: Check of the formula in the case $n = 1$.*Source: own material.*

Degree k	$\mathbf{H_k(S(T\mathbb{CP}^1);\mathbb{Z})}$
0	$\mathbb{Z}$
1	$\mathbb{Z}/2$
2	0
3	$\mathbb{Z}$

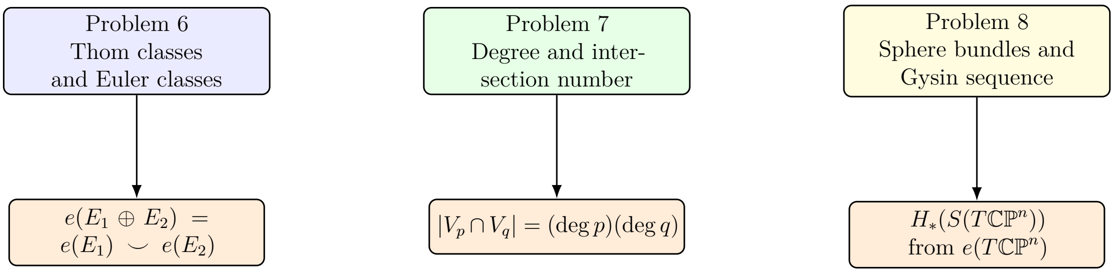


Figure 12: Conceptual summary of Problems 6–8.*Źródło: opracowanie własne.*